

co2amp

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2026-03-26

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Chapter 1

General notes

1.1 Program capabilities

1. Ultrashort pulse amplification in CO₂ active medium
 - Rotational numbers $J = 0 \dots 79$
 - Regular, hot, and sequence bands
 - Isotopic CO₂
2. Molecular dynamics
 - Realistic pumping
 - Collisional relaxation processes
 - Stimulated transitions
 - Independent consideration of active medium regions at different elongations from the optical axis
3. Diffraction-based beam propagation
 - Beam manipulation with common optical elements
 - Arbitrary optical configurations
4. Linear dispersion and non-linear effects in optical materials
 - Pulse chirping
 - Kerr lensing
 - Self-phase modulation
5. Advanced optics
 - Chirped-pulse amplification
 - Spectral filtering
 - Trains of pulses
 - Staging (program output as an input for the next stage)
6. User's interface
 - Easy specification of parameters
 - Graphical output
 - Project save/recall

1.2 Availability, Tools, and Third Party Components

The simulation core **co2amp** and the user's interface shell **co2am+** are written in the C++ programming language. **co2am+** utilizes the QT library (<http://qt.io>), and QT Creator, a component of the QT project, is employed as the development environment. Windows executables are compiled using the MinGW compiler, which is part of the open-source QT distribution. The code is hosted on GitHub (<https://github.com/polyanskiy/co2amp>) and is freely available for use, modification, and redistribution under the GNU General Public License (GPL v.3) (<https://www.gnu.org/licenses/gpl-3.0.html>). A binary package is available as a Windows installer, containing pre-compiled executables, documentation, templates, and examples at <https://github.com/polyanskiy/co2amp/releases/>. The project leverages cross-platform libraries, facilitating compilation on other platforms (MacOS, Linux). **co2amp** relies on three third-party components: gnuplot, 7-zip, and HDF5, available at <http://www.gnuplot.info/>, <https://www.7-zip.org/>, and <https://www.hdfgroup.org/solutions/hdf5/>, respectively. These components must be installed separately. The Windows installer is created using the Nullsoft Scriptable Install System (NSIS, <https://nsis.sourceforge.io/>), representing the only platform-specific component of the project. The documentation is primarily written in L^AT_EX (<http://www.latex-project.org>) using the Overleaf online editor and compiler (<https://www.overleaf.com/>). YAML and HDF5 file formats are adopted for specifying input parameters and storing output field information, respectively.

1.3 Acknowledgements

Viktor Platonenko from Moscow State University (Russia) provided a Mathcad code for pulse amplification in the CO₂ active medium, which served as the starting point for developing the **co2amp** program. Dr. Platonenko also offered valuable input during the early stages of the work on **co2amp**.

Chapter 2

Basic concepts

2.1 `co2amp` and `co2am+`

`co2amp` is a terminal program for simulating the propagation of ultrashort pulses through an arbitrary cylindrically symmetric optical system, which may include CO₂ amplifiers. It uses specially formatted text files and command line arguments as inputs and produces tabulated output files and a binary file containing complete information on the output field. Although `co2amp` can be used independently, the workflow is greatly simplified by the `co2am+` graphical user interface.

`co2am+` is a graphical user interface that streamlines the management of multiple input and output files, as well as calculation parameters, by maintaining an organized and easily navigable working environment. `co2am+` can save and restore the entire project file structure together with command line parameters in a single compressed project file (extension `.co2`).

2.2 Projects

A `co2amp` project is defined by: the initial `pulse(s)`, the optical `layout`, the configuration of all `optics` (including amplifiers), and the calculation parameters (e.g., the calculation grid).

For each `optic` in the `layout`, the temporal and spatial structure of the `pulse` is saved and can be inspected both graphically and as tabulated data.

`co2am+` stores all project inputs and most outputs in a single compressed `.co2` file. The complete output pulse field (complex $E(t, \rho)$ at every node of the space-time calculation grid) can additionally be saved as a binary HDF5 file (extension `.pulse`) and used as an input for another project. An example of the input file structure of a `.co2` project in the `co2am+` interface is shown in Fig. 2.1.

2.3 Pulse, Layout, and Optic

A `pulse` is a complex electric-field envelope defined on the space-time calculation grid. A project can include one or more input pulses, each defined in a separate YAML (`.yaml`) file. A pulse can be specified either by referencing an output from another project (a `.pulse` file) or by explicitly defining the spatial and temporal profile in the YAML file.

A `layout` is a sequence of infinitely thin `optics` separated by free space. Pulses propagate freely between `optics`. Each project has exactly one `layout`, defined in a YAML file that specifies the order of `planes` and the distances between them.

An `optic` is a system element that modifies the pulse as it passes through. Several optic types are described later. For example, an ideal thin *Lens* applies a radial-coordinate-dependent phase, changing the

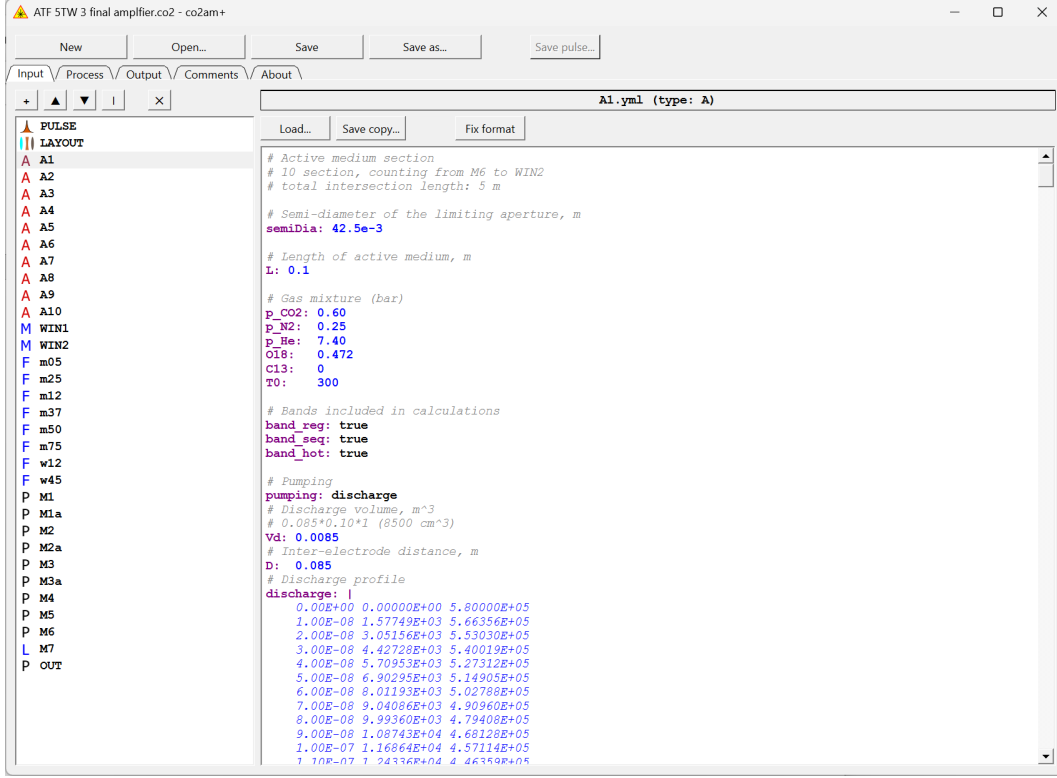


Figure 2.1: "Input" tab of the **co2amp+** user interface program. YAML files specifying the pulse, layout, and optics are listed on the left. The content of the selected file is displayed and can be edited in the editor on the right.

beam divergence. Each **optic** is specified in a separate YAML file. The same **optic** may appear multiple times in a **layout**, as in a laser cavity¹.

co2amp supports seven optic types, listed in Table 2.1.

2.4 Field and Fourier-transform conventions

This section summarizes the definitions and sign conventions used in **co2amp** for the complex optical field, Fourier transforms, and carrier-frequency handling. These conventions are used in all models that manipulate the field in either the time domain (e.g., amplification) or the frequency domain (e.g., diffraction, spectral filtering, dispersion, chirping).

Complex field envelope and physical field

The propagated quantity in **co2amp** is the complex field envelope $E(t, \rho)$, where t is the pulse time (fast time frame) and ρ is the radial coordinate. The physical (real) electric field is represented as

$$E_{\text{phys}}(t, \rho) = \Re\{E(t, \rho)e^{-2\pi i\nu_0 t}\}, \quad (2.1)$$

where ν_0 is the user-specified center frequency of the calculation grid.

¹Internally, **co2amp** also uses the concept of a **plane**. A **plane** is a **layout** element that appears only once. Each **plane** references one **optic**. Conceptually, a **plane** is a placeholder for an **optic**.

Table 2.1: Types of Optics

<i>Type ID</i>	<i>Name</i>	<i>Description</i>
A	<i>Active medium</i>	A CO ₂ amplifier section.
P	<i>Probe</i>	A passive surface. May be used as a limiting aperture.
F	<i>Spatial filter</i>	An optic with coordinate-dependent transmission.
S	<i>Spectral filter</i>	An optic with frequency-dependent transmission.
L	<i>Lens</i>	An ideal thin lens.
M	<i>Material</i>	A material layer introducing dispersion and/or absorption (linear and/or nonlinear).
C	<i>Chirper</i>	An optic that applies chirp to the pulse (typically a stretcher or compressor).

Fourier transform convention

The Fourier transform operator $\mathcal{F}$ used throughout the documentation (and implemented by the FFT routines) is defined by

$$\widehat{E}(\nu, \rho) = \mathcal{F}(E(t, \rho)) = \int_{-\infty}^{\infty} E(t, \rho) e^{-2\pi i \nu t} dt, \quad (2.2)$$

and the inverse transform by

$$E(t, \rho) = \mathcal{F}^{-1}(\widehat{E}(\nu, \rho)) = \int_{-\infty}^{\infty} \widehat{E}(\nu, \rho) e^{+2\pi i \nu t} d\nu. \quad (2.3)$$

Accordingly, a monochromatic component $E(t, \rho) \propto e^{-2\pi i \nu t}$ appears at positive frequency ν in $\widehat{E}(\nu, \rho)$.

Carrier frequency and frequency shifting

The calculation grid is centered at frequency ν_0 , specified by the user. Each input **pulse** is specified by its carrier frequency ν_c (parameter **freq** in the pulse YAML file). Internally, the envelope $E(t, \rho)$ is stored with respect to the grid carrier ν_0 .

To represent a pulse defined with carrier ν_c on the grid carrier ν_0 , **co2amp** applies a phase ramp:

$$E_{(\nu_0)}(t, \rho) = E_{(\nu_c)}(t, \rho) \exp[-2\pi i (\nu_0 - \nu_c) t]. \quad (2.4)$$

The inverse operation (converting an envelope referenced to ν_0 back to ν_c) is obtained by changing the sign in the exponent.

Detuning sign convention in the active-medium model

During pulse interaction with an active medium, detuning of a ro-vibrational transition j is defined as $\nu_c - \nu_j$, where ν_j is the line-center frequency of the transition and ν_c is the pulse carrier frequency for the interaction step. With the field convention above, the free polarization evolution includes the phase factor

$$\exp[-2\pi i (\nu_c - \nu_j) \Delta t]. \quad (2.5)$$

Practical note on spectra

Depending on FFT bin ordering, spectra may be stored in wrapped order. For plotting and analysis it is often convenient to apply an FFT-shift operation so that the grid center frequency ν_0 appears in the middle of the array, yielding an axis of the form $\nu = \nu_0 + \Delta\nu (n - N_t/2)$.

2.5 Calculation Grid

A **pulse** is defined on a two-dimensional space-time calculation grid that moves with the pulse. The grid is primarily specified via **co2amp** command line arguments. The only exception is the maximum radial coordinate, which is determined by the semi-diameter of the clear aperture of the current **optic** and therefore may vary along the **layout**.

Time grid

The pulse time grid is defined by the user-specified parameters $t_{\min}$, $t_{\max}$, and the number of time points N_t . The time step is

$$\Delta t = \frac{t_{\max} - t_{\min}}{N_t}. \quad (2.6)$$

Time samples are taken at the midpoints of the bins,

$$t_n = t_{\min} + \Delta t \left(n + \frac{1}{2}\right), \quad n = 0, 1, \dots, N_t - 1. \quad (2.7)$$

The time window and N_t define the corresponding frequency grid used by FFT-based operations:

$$\Delta \nu = \frac{1}{t_{\max} - t_{\min}}, \quad \nu_{\max} - \nu_{\min} = \frac{1}{\Delta t}. \quad (2.8)$$

Thus, the time window must be long enough to provide adequate frequency resolution, while Δt must be small enough to cover the full spectral region of interest.

The grid center frequency ν_0 is also specified by the user and defines the reference carrier used for storing the field envelope internally (see Sec. 2.4).

Radial grid

The radial grid is defined by the number of radial points N_ρ and the maximum radius $\rho_{\max}$. In **co2amp**, $\rho_{\max}$ is set by the clear aperture of the current **optic**:

$$\rho_{\max} = \text{semiDia}. \quad (2.9)$$

Each **optic** therefore has its own $\rho_{\max}$, and the pulse field is represented on the corresponding radial grid at that plane.

The radial step is

$$\Delta \rho = \frac{\rho_{\max}}{N_\rho}, \quad (2.10)$$

and samples are taken at bin midpoints,

$$\rho_x = \Delta \rho \left(x + \frac{1}{2}\right), \quad x = 0, 1, \dots, N_\rho - 1. \quad (2.11)$$

This midpoint sampling is consistent with the numerical integration over concentric rings used in **co2amp** (e.g., when computing pulse energy by integrating intensity over the optic aperture area).

Field normalization and intensity

In **co2amp**, the complex envelope $E(t, \rho)$ is normalized such that the optical intensity is

$$I(t, \rho) = 2h\nu_c |E(t, \rho)|^2. \quad (2.12)$$

With this normalization, $2|E|^2$ is the photon flux density.

Practical guidance

The number of nodes in both dimensions of the pulse grid is always a power of two, enabling efficient Fast Fourier Transform (FFT) algorithms. Increasing grid resolution generally improves accuracy but increases both computation time and memory usage (approximately proportional to $N_t N_\rho$). A practical workflow is to start with a smaller grid and increase the resolution until the output no longer changes significantly.

The pulse time frame is used for all **pulse**-related calculations (interaction with **optics** and free-space propagation) and for fast processes in some **optics** (e.g., stimulated transitions and rotational relaxation in an *Active Medium*). Processes that are much slower than the pulse duration (e.g., pumping and vibrational relaxation) are modeled in a separate laboratory time frame. The time step of the laboratory time frame is also specified via a **co2amp** command line argument.

Choosing an appropriate calculation grid is essential for obtaining accurate and efficient simulations.

2.6 Units

co2amp uses SI base units without prefixes (meters, seconds, amperes; not centimeters, nanoseconds, kiloamperes) for input, output, and internal calculations. **co2amp+** allows changing units used for graphical display on the "Output" tab (Fig. 2.2). When numerical data are accessed via [Right-click on a plot] – [Copy raw data],

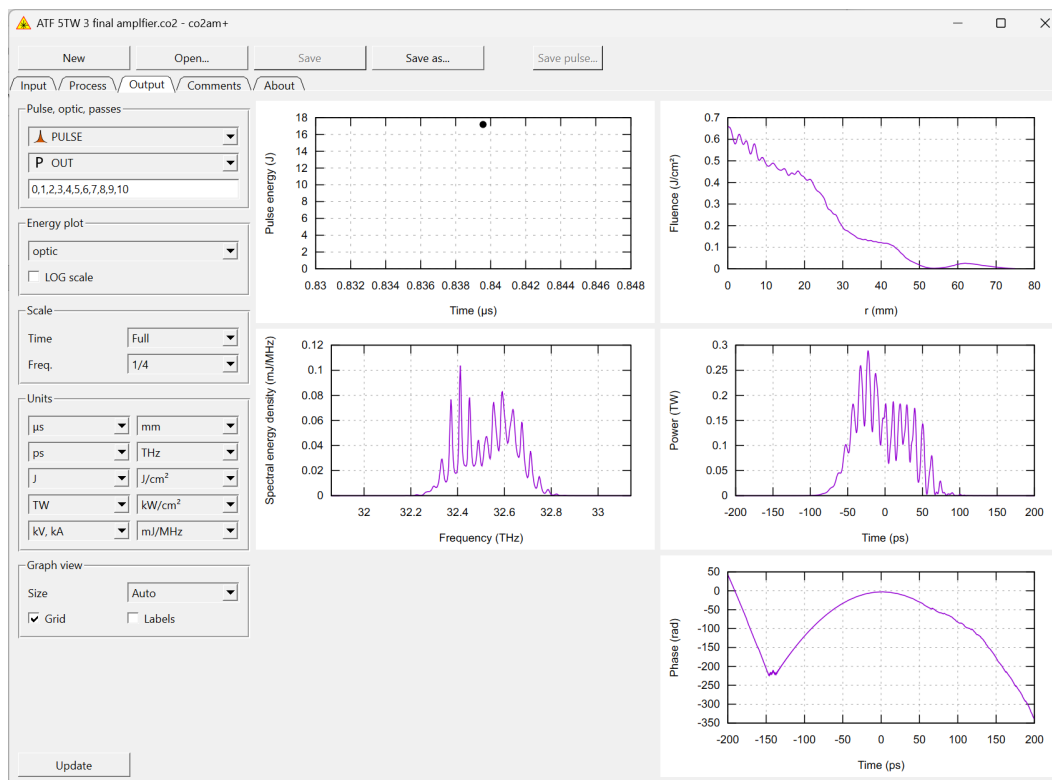


Figure 2.2: "Output" tab of the **co2amp+** user interface program. Controls on the left allow selecting the data to display and fine-tuning the look of the plots.

the units are always stored in the prefix-less SI form.

2.7 Program Output

Program output includes the temporal and spatial structure of each **pulse** at every **optic** within the **layout**. Temporal (and spectral) profiles are integrated over the optic aperture area, while spatial profiles are integrated over the pulse time window.

In the **co2am+** "Output" tab, users can choose a **pulse** and an **optic** to display (Fig. 2.2). If the selected **optic** is used multiple times in the **layout**, the corresponding pass number can be selected. Pulse energy can be displayed either for a selected optic across passes, or for all optics in the layout.

Some optic types provide additional type-specific output. For an *Active Medium*, this includes gain, discharge profile, population dynamics, and pumping-energy partition (fractions of discharge energy contributing to excitation of laser levels, heating of translational modes, and ionization). Output for an optic of type *Probe* includes the phase of the optical field at the beam center.

2.8 "Comments" and "About" Tabs of co2am+

The "Comments" tab in **co2am+** provides an editable text box where users can enter project notes. These comments are stored as part of the project in the `.co2` file.

The "About" tab contains information about the versions of **co2amp** and **co2am+**, including links to the license and the documentation, author contact information, and a suggested citation format.

Chapter 3

Elements of a project

A project in **co2amp** must include the following elements, each specified in separate input YAML (‘.yaml’) files:

1. One or more **pulses**
2. One or more **optics**
3. One **layout**

Each element is detailed in its dedicated YAML file¹. The last **optic** in the **layout** must be of type **P** (*Probe*).

The subsequent sections provide brief descriptions of each of these elements and the models associated with them. For a comprehensive understanding, refer to the templates, example files, and the comments within them.

3.1 Pulse

Unless utilizing the output of another project (a ‘pulse’ file) as input, both the temporal and spatial shape of the input **pulse** must be defined in a corresponding YAML (‘.yaml’) file. The **pulse** is assumed to be transform-limited, meaning it has no initial chirping. Specifications such as the **pulse** energy, central frequency, and injection time are also required. The injection time denotes the time-delay between the zero moment of the laboratory time frame (“slow” time frame) and the injection of a **pulse** into the optical system (the first **optic** in the **layout**). An example of a **pulse** configuration file is provided below.

```
#=====
# PULSE.yaml from 'examples/00 simple propagation.co2' project

t_in: 0
E: 1e-3
freq: 32.5e12

beam: GAUSS
w: 3e-3

pulse: GAUSS
fwhm: 2e-12
```

¹**co2amp** additionally requires an input file ‘config_files.yaml’ that enumerates all input YAML files and the types of corresponding elements. **co2amp+** automatically generates this file.

```
#=====
```

This file specifies a 2 ps (FWHM) transform-limited Gaussian pulse with a $w = 3$ mm Gaussian beam profile, 1 mJ energy, and a 32.5 THz central frequency, injected into the system at $t_{\text{in}} = 0$. Several pre-defined beam and pulse profile options are available, such as **GAUSS**, **FLATTOP**, **SUPERGAUSS4**, **SUPERGAUSS6**, etc. Alternatively, a **FREEFORM** option allows for the specification of an arbitrary shape through a tabulated numerical profile (refer to the 'pulse.yml' template for details).

3.2 Layout

3.2.1 Configuration

The **layout** configuration defines the sequence of **optics** and the distances between them in the optical system. Below is an example of a simple **layout** configuration file:

```
#=====
# LAYOUT.yml from 'examples/00 simple propagation.co2' project

- go: P1 >> 3 >> P2
  times: 1
#=====
```

In this example, the system consists of two **optics**, P1 and P2, separated by 3 meters of free space. The pulses pass through the system once. If the **times** value is greater than 1, a pulse after passing through P2 will return to P1, and the propagation through the system will repeat for the specified number of times. A **layout** configuration file can contain several such "go-times" sequences. Below is an example of a **layout** configuration for a more complex system:

```
#=====
# LAYOUT.yml from 'examples/ATF 5 TW/ATF 1 regen.co2' project

- go: str >> COU1
  times: 1

- go: 0.45 >> i >> 0.90 >> GE >> 0.25 >> w >> WIN1 >> w >> 0.45 >> AM1 >> 0.40 >> AM2 >> 0.45
>> w >> WIN2 >> w >> 0.10 >> MIR >> m >> 0.10 >> w >> WIN2 >> w >> 0.45 >> AM2 >> 0.40 >> AM1
>> 0.45 >> w >> WIN1 >> w >> 0.25 >> GE >> 0.90 >> i >> 0.45 >> COU2
  times: 15

- go: 0.45 >> i >> 0.90 >> GE >> 0.25 >> w >> WIN1 >> w >> 0.45 >> AM1 >> 0.40 >> AM2 >> 0.45
>> w >> WIN2 >> w >> 0.10 >> MIR >> m >> 0.10 >> w >> WIN2 >> w >> 0.45 >> AM2 >> 0.40 >> AM1
>> 0.45 >> w >> WIN1 >> w >> 0.25 >> OUT
  times: 1
#=====
```

3.2.2 Dealing with Long Optical Elements

In the **co2amp** model, **optics** are considered infinitely thin. For long **optics**, such as an *Active Medium*, the model calculates the field modification accumulated by a **pulse** as it propagates through the **optic** and then applies this modification as if it occurred instantaneously. However, this approach might not be accurate if the actual optical element is lengthy and the **pulse** changes significantly while propagating through it, thereby interacting differently with various parts of the **optic**. The model's accuracy can be improved by dividing long elements into shorter sub-sections.

Fig. 3.1 illustrates an example of a 2-meter long layout with a meter-long active medium in the middle. In one scenario, shown in Fig. 3.1a, we first propagate the pulse to the midpoint of the amplifier section, then apply the amplification accumulated over 1 meter, and finally propagate the pulse to the last optic. The corresponding layout configuration is:

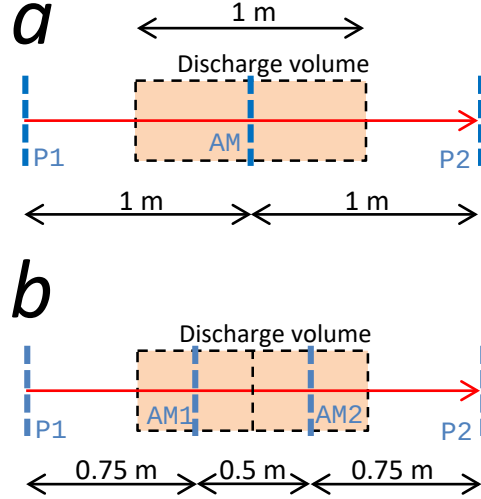


Figure 3.1: Example of layout configuration for a long optic (in this case, an *Active Medium*). a) The *Active Medium* is represented by a single optic. b) The *Active Medium* is split into two shorter sections.

```
#####
# long amplifier
- go: P1 >> 1 >> AM >> 1 >> P2
  times: 1
#####
```

Alternatively, the active medium can be represented by two 0.5-meter sections, as shown in Fig. 3.1b. The corresponding layout is:

```
#####
# long amplifier divided into two shorter sections
- go: P1 >> 0.75 >> AM1 >> 0.5 >> AM2 >> 0.75 >> P2
  times: 1
#####
```

By splitting a long amplifier into shorter sections, the population dynamics within each amplifier section is modeled more accurately, leading to a more realistic representation of the active medium.

3.2.3 Modeling of Pulse Propagation Between Optics

Consider free-space wave propagation between plane-parallel surfaces S' and S , separated by distance z , as illustrated in Fig. 3.2 for a system with cylindrical symmetry. According to the Huygens-Fresnel principle, the field E at a point on plane S is defined as a superposition of secondary waves emitted from every point

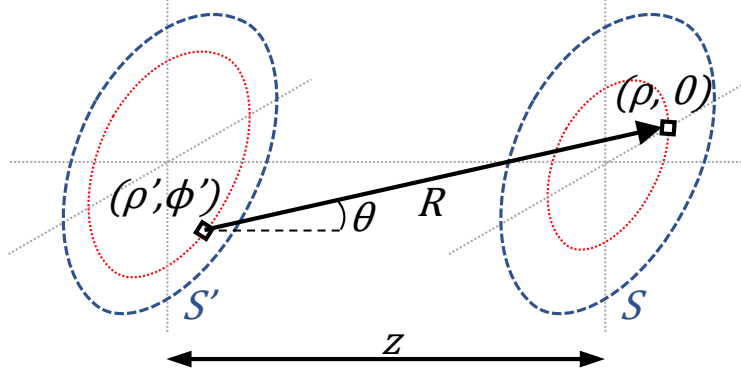


Figure 3.2: Application of the Huygens-Fresnel principle to beam propagation from plane S' to plane S in a system with cylindrical symmetry.

of plane S' [1]. This can be expressed in the case of cylindrical symmetry as [2, 3]:

$$E(\rho) = -\frac{i}{\lambda} \int_{\rho'=0}^{\infty} E'(\rho') \int_{\phi'=0}^{2\pi} \frac{e^{ikR}}{R} K d\phi' \rho' d\rho' \quad (3.1a)$$

$$R = \sqrt{\rho^2 + \rho'^2 + z^2 - 2\rho\rho' \cos \phi'} \quad (3.1b)$$

$$K = \cos \theta = \frac{z}{R} \quad (3.1c)$$

where λ is the wavelength, $k = 2\pi/\lambda$ the wavenumber, and K the obliquity factor as it appears in Rayleigh-Sommerfeld diffraction theory.

Since the field on the output plane S does not depend on the angular coordinate ϕ , $\phi = 0$ is chosen for the simplification of Eq. 3.1.

Direct numerical integration of Eq. 3.1, with $O(N^3)$ complexity, is very time-consuming. Therefore, an approximation is usually employed to accelerate computations. The most well-known approximation is Fresnel diffraction, which assumes:

$$K \approx 1$$

$$R \approx \begin{cases} z & \text{(denominator)} \\ z \left(1 + \frac{\rho^2 + \rho'^2 - 2\rho\rho' \cos \phi'}{2z^2} \right) & \text{(exponent)} \end{cases} \quad (3.2)$$

where "denominator" and "exponent" indicate the position of the R variable in Eq. 3.1a.

Substituting Eq. 3.2 into Eq. 3.1a and using the formula

$$\int_0^{2\pi} e^{\pm ia \cos \phi} d\phi = 2\pi J_0(a) \quad (3.3)$$

where J is the Bessel function, we obtain the expression for Fresnel diffraction with cylindrical symmetry:

$$E(\rho) \approx -\frac{2\pi i e^{ik\left(z + \frac{k\rho^2}{2z}\right)}}{\lambda z} \int_0^{\infty} E'(\rho') e^{i\frac{k\rho'^2}{2z}} J_0\left(\frac{k\rho\rho'}{z}\right) \rho' d\rho' \quad (3.4)$$

co2amp supports both Rayleigh-Sommerfeld (Eq. 3.1) and Fresnel (Eq. 3.4) based propagation methods. Users can also choose to ignore the **pulse** evolution during free-space propagation.

Eqs. 3.1 and 3.4 assume monochromatic light, which is not the case for ultrashort pulses that possess a non-negligible bandwidth. Therefore, in **co2amp**, propagation is calculated in the frequency domain: Eqs. 3.1 or 3.4 are applied to the Fourier-transformed field at each node of the frequency calculation grid. Afterward, an inverse Fourier transform is used to return to the time domain.

3.3 Optic Type A: *Active Medium*

The *Active Medium* is the most complex type of **optic** that can be utilized in a **co2amp** project. Detailed models used for simulating molecular dynamics and **pulse** amplification are described in a dedicated Chapter 4.

A configuration file for an **optic** of type A must include specifications of the gas mixture, pumping mechanism, and laser transitions considered in the simulations. An example of such a configuration file is provided below:

```
#=====
# AM1.yml from 'examples/ATF 5 TW/ATF 3 final amplifier.co2' project

# Semi-diameter of the limiting aperture, m
semiDia: 45e-3

# Length of active medium, m
L: 0.57

# Gas mixture (bar)
p_CO2: 0.50
p_N2: 0.25
p_He: 7.50
O18: 0.472
C13: 0
T0: 300

# Bands included in calculations
band_reg: true
band_seq: true
band_hot: true

# Pumping
pumping: discharge
# Discharge volume, m^3
Vd: 0.0085
# Inter-electrode distance, m
D: 0.085
# Discharge profile
discharge: |
    0.00E+00 0.00000E+00 5.80000E+05
    1.00E-08 1.97186E+03 5.66356E+05
    2.00E-08 3.81445E+03 5.53030E+05
    3.00E-08 5.53410E+03 5.40019E+05
    4.00E-08 7.13691E+03 5.27312E+05
    ...
#=====
```


The composition of the active medium, including isotopic enrichment of carbon dioxide, and the initial temperature are specified under the "Gas mixture (bar)" section.

For discharge pumping, the geometry of the discharge and its temporal profile are required. In the case of optical pumping, the wavelength, absorption cross-section, and the temporal profile of the pumping pulse must be provided.

The 'optic A (discharge pumped CO2 amplifier).yaml' and 'optic A (optically pumped CO2 amplifier).yaml' template files contain detailed information on the configuration file format and can be referred to for further guidance.

3.4 Optic Type P: *Probe*

A *Probe* is a passive type of **optic**. It does not alter the field that fits within its semi-diameter. This can be expressed mathematically as:

$$E(t, \rho) = E'(t, \rho) \quad (3.5)$$

where $E'(t, \rho)$ and $E(t, \rho)$ represent the field before and after passing through an **optic**, respectively.

However, a *Probe* **optic** can serve as a limiting aperture, exhibiting zero transmittance for $\rho > \text{semiDia}$. The sole configuration parameter for an **optic** of type P is its semi-diameter. An example of a configuration file for a *Probe* with a 25 mm semi-diameter is shown below:

```
#####
# probe

semiDia: 25e-3
#####
```

3.5 Optic Type F: *Spatial Filter*

A *Spatial Filter* applies a specified coordinate-dependent transmittance function to a **pulse**:

$$E(t, \rho) = E'(t, \rho) \sqrt{\mathcal{T}(\rho)} \quad (3.6)$$

where $\mathcal{T}(\rho)$ is the transmittance function, as defined in the configuration file.

An example configuration for a *Spatial Filter* is shown below:

```
#####
# spatial filter

semiDia: 25e-3

filter: SIN
R: 10e-3
w: 10e-3
#####
```

For more details and configuration options, refer to the 'optic F (spatial filter).yaml' template file.

3.6 Optic Type S: *Spectral Filter*

A *Spectral Filter* applies a specified frequency-dependent transmittance function to a pulse:

$$\begin{aligned}\hat{E}'(\nu, \rho) &= \mathcal{F}(E'(t, \rho)) \\ \hat{E}(\nu, \rho) &= \hat{E}'(\nu, \rho) \sqrt{\mathcal{T}(\nu)} \\ E(t, \rho) &= \mathcal{F}^{-1}(\hat{E}(\nu, \rho))\end{aligned}\tag{3.7}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier transform and the inverse Fourier transform, respectively, ν is the frequency, and $\mathcal{T}(\nu)$ is the transmittance function as defined in the configuration file.

An example configuration for a *Spectral Filter* is provided below:

```
#=====
# spectral filter

semiDia: 25e-3

filter: FREEFORM
form: |
    32.0e12 1.0
    32.1e12 0.9
    32.2e12 0.7
    32.3e12 0.5
    32.4e12 0.3
    32.5e12 0.0
    32.6e12 0.3
    32.7e12 0.5
    32.8e12 0.7
    32.9e12 0.9
    33.0e12 1.0
#=====
```

For further details and configuration options, refer to the 'optic S (spectral filter).yaml' template file.

3.7 Optic Type L: *Lens*

A *Lens* functions as a standard optical lens within the system:

$$\begin{aligned}\hat{E}'(\nu, \rho) &= \mathcal{F}(E'(t, \rho)) \\ \hat{E}(\nu, \rho) &= \hat{E}'(\nu, \rho) \exp\left(-\frac{ik\rho^2}{2F}\right) \\ E(t, \rho) &= \mathcal{F}^{-1}(\hat{E}(\nu, \rho))\end{aligned}\tag{3.8}$$

where $k = \frac{2\pi\nu}{c}$ is the wave number (c is the speed of light) and F is the focal length of the lens.

The calculation is performed in the frequency domain to ensure that the effective focal length remains consistent across all frequencies in the pulse spectrum.

An example configuration for a lens with a 1-meter focal length is shown below:

```
#=====
# lens (F = 1 m)
```

semiDia: 25e-3

F: 1.0

#=====

3.8 Optic Type M: *Material*

In cases of oblique incidence, the effective intensity I_{eff} is reduced and the propagation distance in the material (effective thickness) Θ_{eff} is automatically adjusted based on the incidence angle θ_i and the refractive index n :

$$\begin{aligned}\theta_r &= \arcsin\left(\frac{\sin \theta_i}{n_0}\right) \\ I_{\text{eff}} &= I \frac{\cos \theta_i}{\cos \theta_r} \\ \Theta_{\text{eff}} &= \frac{\Theta}{\cos \theta_r}\end{aligned}\tag{3.9}$$

where I and Θ are the intensity before the **optic** and the actual thickness of the material, respectively, and θ_r is the refraction angle.

Linear Dispersion and Absorption

$$\begin{aligned}\hat{E}'(\nu, \rho) &= \mathcal{F}(E'(t, \rho)) \\ \hat{E}(\nu, \rho) &= \hat{E}'(\nu, \rho) \exp(i\Delta\phi) \sqrt{\exp(-\alpha_0 \Theta_{\text{eff}})} \\ E(t, \rho) &= \mathcal{F}^{-1}(\hat{E}(\nu, \rho))\end{aligned}\tag{3.10}$$

where the phase shift $\Delta\phi$ is defined as:

$$\Delta\phi = 2\pi \int_0^\nu (\nu' - \nu_c) \frac{dt}{d\nu'} d\nu',\tag{3.11}$$

$$\frac{dt}{d\nu'} = \frac{\Theta_{\text{eff}}}{c} \frac{dn_g}{d\nu'},\tag{3.12}$$

with c as the speed of light, n_g as the group index of refraction, and ν_c as the central frequency. The dispersion formulas used for calculating n_g are given in Appendix C.

Nonlinear Interaction

$$\begin{aligned}E(t, \rho) &= E'(t, \rho) \exp\left(2\pi i \nu_c \frac{\Theta_{\text{eff}}}{c} n_2 I_{\text{eff}}(t, \rho)\right) \\ I_{\text{eff}}(t, \rho) &= 2h\nu_c (E'(t, \rho))^2 \frac{\cos \theta_i}{\cos \theta_r}\end{aligned}\tag{3.13}$$

where n_2 is the nonlinear refractive index, h is Planck's constant, and $I(t, r)$ is the field intensity. Numerical values of n_2 used in the program are given in Appendix C.

Configuration example for a *Material optic*:

#=====

material

semiDia: 25e-3

```

material: NaCl
thickness: 100e-3
tilt: 0
slices: 10
#=====

```

Currently supported materials include AgBr, AgCl, BaF₂, CdTe, CsI, GaAs, Ge, IRG22 (AMTIR1), IRG24, IRG25, KBr, KCl, KRS5, NaCl, NaF, Si, SiO₂, ZnS, ZnSe, and air. An arbitrary n_2 can be specified in the configuration file, with a predefined value used otherwise (see Appendix C). To enhance accuracy, the *Material optic* can be divided into several layers. A split-step method is employed for calculating linear and nonlinear interactions with a layer: first, a nonlinear interaction with a half-layer is calculated, followed by a full-layer linear interaction, and then a half-layer nonlinear interaction again.

3.9 Optic Type C: *Chirper*

A *Chirper* introduces a chirp to a pulse and is typically used to model a stretcher or compressor.

$$\begin{aligned}
\hat{E}'(\nu, \rho) &= \mathcal{F}(E'(t, \rho)) \\
\hat{E}(\nu, \rho) &= \hat{E}'(\nu, \rho) \exp(i\Delta\phi) \\
E(t, \rho) &= \mathcal{F}^{-1}(\hat{E}(\nu, \rho))
\end{aligned} \tag{3.14}$$

where

$$\Delta\phi = 2\pi \int_0^\nu (\nu' - \nu_c) \frac{dt}{d\nu'} d\nu' \tag{3.15}$$

is the phase shift, ν_c is the central frequency, and $\frac{d\nu}{dt}$ is the chirp rate.

In the case of linear chirp, the chirp rate is constant, and Eq. 3.15 simplifies to:

$$\begin{aligned}
\Delta\phi &= 2\pi \int_0^\nu \frac{\nu' - \nu_c}{\mathcal{C}} d\nu' = \pi \frac{(\nu - \nu_c)^2}{\mathcal{C}} \\
\mathcal{C} &= \frac{d\nu}{dt}
\end{aligned} \tag{3.16}$$

Positive chirp rate corresponds to a red chirp of the pulse.

An example configuration for a *Chirper* with linear chirp is shown below:

```

#=====
# stretcher (positive chirp <=> red chirp)

semiDia: 25e-3

chirp: LINEAR
c: 3.5e21
#=====

```

Chapter 4

Modeling of processes in CO₂ amplifiers

4.1 Basics of the molecular spectroscopy of CO₂

4.1.1 Isotopologues of CO₂ and their nomenclature

The **co2amp** model includes twelve isotopologues of CO₂ with different combinations of stable isotopes of carbon (¹²C and ¹³C) and oxygen (¹⁶O, ¹⁷O and ¹⁸O). A commonly used three-digit notation designates the isotopologues of carbon dioxide, where each digit represents the isotope of an atom in the molecule in the order oxygen–carbon–oxygen, corresponding to the last digit of the isotope’s mass number. In this notation, the digits *2* and *3* represent ¹²C and ¹³C, respectively, while the digits *6*, *7* and *8* represent ¹⁶O, ¹⁷O and ¹⁸O, respectively. Thus, *626*, for instance, denotes a CO₂ molecule with the natural isotopic composition ¹⁶O-¹²C-¹⁶O, and *638* stands for the ¹⁶O-¹³C-¹⁸O isotopologue.

4.1.2 Vibrational levels of CO₂ molecule

Laser transitions

Laser transitions in CO₂ amplifiers occur between rotational sublevels of vibrational levels within the electronic ground state of the molecule. The primary laser transitions (the “regular bands”) occur between the first excited state of the antisymmetric stretching mode and combination states involving the first excited symmetric stretching mode and the second excited bending mode. However, transitions between higher-lying energy levels are also possible and may contribute to the overall gain of the amplifier. Multiple vibrational energy levels are therefore included in the **co2amp** amplification model. Figure 4.1 shows the lower-lying vibrational levels that are most important for the molecular dynamics, together with the laser transitions included in the **co2amp** amplification model.

Below, we summarize the relevant molecular spectroscopy and then describe the model for population dynamics and stimulated emission.

Vibrational level nomenclature

A standard scheme of vibrational level nomenclature uses the $\nu_1 \nu_2^{l[e/f]} \nu_3$ notation, where ν_1 , ν_2 , and ν_3 are the numbers of quanta in the symmetric stretch, bending, and antisymmetric stretch vibrational modes, respectively, and l is the vibrational angular momentum quantum number of the doubly degenerate bending vibration. States with $l \neq 0$ are further split into two sub-levels due to the two possible symmetries associated with the bending vibration. A letter *e* or *f* is added to the notation after the value of l to differentiate between these sub-levels.

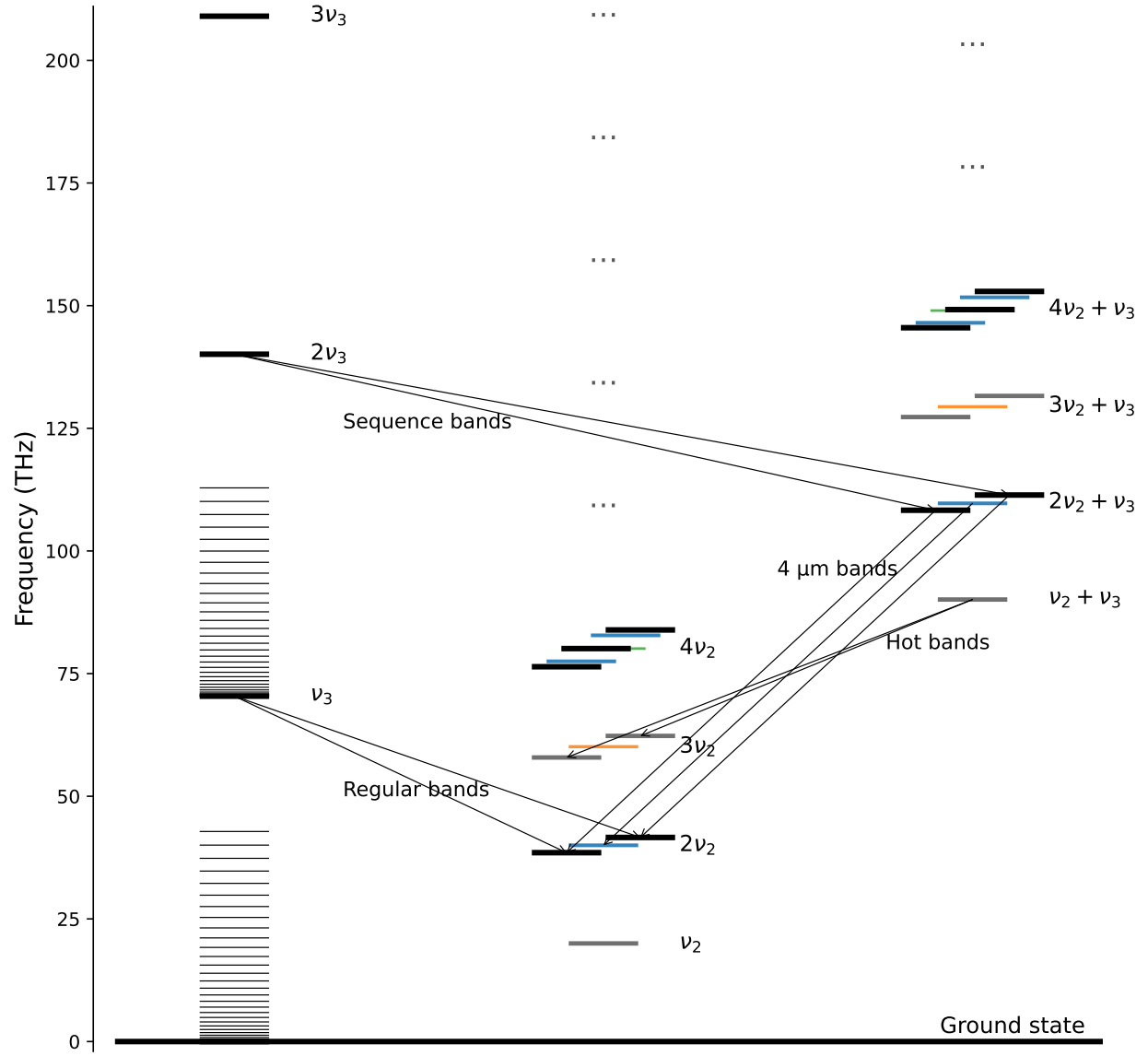


Figure 4.1: Lower vibrational levels of the CO_2 molecule and rotational sublevels of selected levels. To label groups of levels, we specify the number of quanta in the asymmetric stretch mode ν_3 and the combined number of quanta in the Fermi-resonance-coupled symmetric stretch ν_1 and bending ν_2 modes. To simplify the notation, each quantum of ν_1 is represented as $2\nu_2$. Different line colors indicate different values of the vibrational angular momentum l : black for $l = 0$, gray for $l = 1$, blue for $l = 2$, orange for $l = 3$, and green for $l = 4$. The level frequencies correspond to the natural CO_2 isotopologue (626). The laser transitions included in the model are indicated. Rotational sublevels with rotational quantum number J up to 60 are shown only for the ground state and the ν_3 level, for clarity.

In the CO₂ molecule, a strong Fermi coupling exists between the ν_1 and ν_2 vibrations, resulting in mixed states that cannot be directly attributed to a single set of vibrational quantum numbers. We refer to these levels by listing the contributing states in square brackets and adding a Roman numeral subscript indicating a sub-level number. For instance, the lower laser level of the 9.4 μm regular band is denoted as $[1\ 0^0\ 0, 0\ 2^0\ 0]_{\text{II}}$.

In the HITRAN database (see Section 4.1.4), the symmetry of the bending vibration (e or f) is associated with the rotational levels rather than with the vibrational ones, as in the **co2amp** model. Associating the symmetry with the vibrational state makes modeling the energy distribution between ro-vibrational sub-levels more straightforward and transparent. Also, the letter e is associated with the $l = 0$ levels in HITRAN for database consistency.

Furthermore, in HITRAN, the mixed states are labeled by one of the contributing states followed by a sub-level number. The $[1\ 0^0\ 0, 0\ 2^0\ 0]_{\text{II}}$ level, for example, is denoted as $1\ 0^0\ 0\ (2)$ (or $1\ 0\ 0\ 2$ in the raw `.par` files). The letter e is then added to the rotational level labels (e.g., "P 20e").

Vibrational levels and transitions included in the co2amp model

Table 4.1 lists the vibrational levels included in the model. The levels in the table are grouped to include sub-levels arising from angular momentum splitting and Fermi coupling between the ν_1 and ν_2 vibrations.

Table 4.2 lists the supported transitions.

Please note that support for the 4- μm transitions is experimental and is not included in the calculations by default. To enable these transitions, use the `band_4um: true` instruction in the configuration of the active medium. It is also recommended to disable other transitions when enabling this feature.

4.1.3 Rotational sub-levels

Rotational sub-level population in the rotational equilibrium is calculated as

$$N_{rot}^0(J) = z(J) \times s(J) \times N_{vib}. \quad (4.1)$$

Here, N_{vib} is the total population density of the corresponding vibrational level. $z(J)$ is the rotational Boltzmann distribution function defined by:

$$z(J) = \frac{hB}{kT} (2J + 1) \exp\left(-\frac{hB}{kT} J(J + 1)\right) \quad (4.2)$$

where B is the rotational constant, $h = 6.626\ 069\ 57 \times 10^{-34}$ J s, and $k = 1.380\ 648\ 8 \times 10^{-23}$ J/K.

To determine the coefficient $s(J)$, follow these steps:

- If $J < l$:

$$s(J) = 0$$

- If $J \geq l$:

- For symmetric isotopologues (*626*, *727*, *636*, *828*, *737*, and *838*):

- * If parity = g and symmetry = e or parity = u and symmetry = f

$$s(J) = \begin{cases} 2 & \text{for even } J \\ 0 & \text{for odd } J \end{cases}$$

- * If parity = g and symmetry = f or parity = u and symmetry = e

$$s(J) = \begin{cases} 0 & \text{for even } J \\ 2 & \text{for odd } J \end{cases}$$

Table 4.1: Vibrational levels included in the model

#	Level	Parity	Description
0	00 ⁰ 0	<i>g</i>	Ground state
			$\nu_3, 2\nu_3, 3\nu_3$
1	00 ⁰ 1	<i>u</i>	Upper level of regular bands; Optical pumping at 4.3 μm
2	00 ⁰ 2	<i>g</i>	Upper level of sequence bands; Optical pumping at 2.2 μm (asymm. molecules)
3	00 ⁰ 3	<i>u</i>	Optical pumping at 1.44 μm
			$2\nu_2$
4	[1 0 ⁰ 0, 0 2 ⁰ 0] _I	<i>g</i>	Lower levels of 4 μm bands;
5	[1 0 ⁰ 0, 0 2 ⁰ 0] _{II}	<i>g</i>	Lower levels of regular bands (4 & 5);
6	0 2 ^{2e} 0	<i>g</i>	Optical pumping at 2.8 μm
7	0 2 ^{2f} 0	<i>g</i>	
			$3\nu_2$
8	[1 1 ^{1e} 0, 0 3 ^{1e} 0] _I	<i>u</i>	Lower levels of hot bands (8, 9, 10, 11)
9	[1 1 ^{1f} 0, 0 3 ^{1f} 0] _I	<i>u</i>	
10	[1 1 ^{1e} 0, 0 3 ^{1e} 0] _{II}	<i>u</i>	
11	[1 1 ^{1f} 0, 0 3 ^{1f} 0] _{II}	<i>u</i>	
12	0 3 ^{3e} 0	<i>u</i>	
13	0 3 ^{3f} 0	<i>u</i>	
			$\nu_2 + \nu_3$
14	0 1 ^{1e} 1	<i>g</i>	Upper levels of hot bands
15	0 1 ^{1f} 1	<i>g</i>	
			$2\nu_2 + \nu_3$
16	[1 0 ⁰ 1, 0 2 ⁰ 1] _I	<i>u</i>	Upper levels of 4 μm bands;
17	[1 0 ⁰ 1, 0 2 ⁰ 1] _{II}	<i>u</i>	Lower levels of sequence bands (16 & 17)
18	0 2 ^{2e} 1	<i>u</i>	
19	0 2 ^{2f} 1	<i>u</i>	
			$4\nu_2 + \nu_3$
20	[2 0 ⁰ 1, 1 2 ⁰ 1, 0 4 ⁰ 1] _I	<i>u</i>	Optical pumping at 2 μm
21	[2 0 ⁰ 1, 1 2 ⁰ 1, 0 4 ⁰ 1] _{II}	<i>u</i>	
22	[2 0 ⁰ 1, 1 2 ⁰ 1, 0 4 ⁰ 1] _{III}	<i>u</i>	
23	[1 2 ^{2e} 1, 0 4 ^{2e} 1] _I	<i>u</i>	
24	[1 2 ^{2f} 1, 0 4 ^{2f} 1] _I	<i>u</i>	
25	[1 2 ^{2e} 1, 0 4 ^{2e} 1] _{II}	<i>u</i>	
26	[1 2 ^{2f} 1, 0 4 ^{2f} 1] _{II}	<i>u</i>	
27	0 4 ^{4e} 1	<i>u</i>	
28	0 4 ^{4f} 1	<i>u</i>	

– For asymmetric isotopologues (627, 628, 728, 637, 638, and 738):

$$s(J) = 1$$

Table 4.2: Supported vibrational transitions

Band #	Transition	Level #'s
Regular bands		
0	$00^0 1 \rightarrow [10^0 0, 02^0 0]_I$	$1 \rightarrow 4$
1	$00^0 1 \rightarrow [10^0 0, 02^0 0]_{II}$	$1 \rightarrow 5$
Hot bands		
2e	$01^{1e} 1 \rightarrow [11^{1e} 0, 03^{1e} 0]_I$	$14 \rightarrow 8$
2f	$01^{1f} 1 \rightarrow [11^{1f} 0, 03^{1f} 0]_I$	$15 \rightarrow 9$
3e	$01^{1e} 1 \rightarrow [11^{1e} 0, 03^{1e} 0]_{II}$	$14 \rightarrow 10$
3f	$01^{1f} 1 \rightarrow [11^{1e} 0, 03^{1f} 0]_{II}$	$15 \rightarrow 11$
Sequence bands		
4	$00^0 2 \rightarrow [10^0 1, 02^0 1]_I$	$2 \rightarrow 16$
5	$00^0 2 \rightarrow [10^0 1, 02^0 1]_{II}$	$2 \rightarrow 17$
4-μm bands		
6	$[10^0 1, 02^0 1]_I \rightarrow [10^0 0, 02^0 0]_I$	$16 \rightarrow 4$
7	$[10^0 1, 02^0 1]_{II} \rightarrow [10^0 0, 02^0 0]_{II}$	$17 \rightarrow 5$
8e	$02^{2e} 1 \rightarrow 02^{2e} 0$	$18 \rightarrow 6$
8f	$02^{2f} 1 \rightarrow 02^{2f} 0$	$19 \rightarrow 7$

In the above:

- l is the vibrational angular momentum. Rotational levels with $J < l$ are not populated due to angular momentum coupling restrictions.
- The **parity** of the vibrational state is determined by the sum of quanta in the ungerade (u) vibrational modes (v_2 and v_3):

$$\text{parity} = \begin{cases} g & \text{if } v_2 + v_3 \text{ is even} \\ u & \text{if } v_2 + v_3 \text{ is odd} \end{cases}$$

- The **symmetry** refers to the e/f symmetry labels of the rotational-vibrational levels, which are determined by the coupling of rotational angular momentum J and vibrational angular momentum l . The e/f labels are adapted from the HITRAN database.
- For symmetric isotopologues, the statistical weight $s(J)$ is 2 for allowed rotational levels due to nuclear spin statistical weights, and 0 for forbidden levels.
- For asymmetric isotopologues, there are no symmetry restrictions due to the lack of identical nuclei, so all rotational levels with $J \geq l$ are populated, and $s(J) = 1$.

4.1.4 Use of HITRAN Data

Data for the supported transitions are extracted from the HITRAN database [4]. These data include transition frequencies (ν), Einstein coefficients (A), and the rotational quantum numbers of the upper and lower laser levels (J_U and J_L , respectively) for each ro-vibrational transition. The molecular constants, particularly the rotational constant B used in Eq. 4.2, for each vibrational level included in the model, are determined by fitting the HITRAN data as described in Appendix B.

Because not all transitions that could influence amplification are included in HITRAN for all isotopologues, additional transitions are, in some cases, included by calculating their frequencies using the fitted molecular constants and assigning them realistic Einstein coefficients. These coefficients are estimated by assuming that all coefficients in a given vibrational transition are equal for all rotational lines. If no rotational line for a vibrational transition is listed, an educated guess is made based on the A values of other bands and isotopologues. Additional `.par` files, containing only the information needed for calculations, are generated to list these estimated transitions and are included with the `co2amp` distribution.

Transitions derived from the original HITRAN data and additional estimated transitions included in separate `.par` files are detailed in Appendix B.

Note that all `.par` files in the `hitran_data` folder are scanned for supported transitions, and all matches are included in the calculations. It is therefore important to ensure that transitions are not duplicated across different files.

4.2 Molecular Dynamics

4.2.1 Time Frames

Two time frames are employed in the `co2amp` models to simulate processes that typically occur at rates differing by several orders of magnitude:

- **Lab time frame:** This time frame is used to describe relatively slow processes, such as the dynamics of active-medium pumping and vibrational relaxation. A typical time step for the lab time frame is on the order of a nanosecond.
- **Pulse time frame:** This is a fast time frame that moves with the laser pulse. It forms part of the spatio-temporal calculation grid used to represent the pulse. The temporal grid must be sufficiently dense to accurately capture the pulse’s temporal structure and wide enough to ensure an appropriate spectral resolution. Processes such as pulse amplification and rotational relaxation are computed within the pulse time frame.

In certain scenarios—such as amplification of a relatively long chirped pulse in a high- CO_2 -concentration active medium—the practically relevant time step in the lab time frame can become shorter than the duration of the pulse time frame. Such situations are handled by splitting the pulse time frame into sections corresponding to the duration of the lab time step and modeling only the amplification of the corresponding pulse section at any given tick of lab time. Thus, the modeling is effectively performed on both the slow and fast time frames simultaneously.

4.2.2 Temperature model

A 3-temperature model is used to describe the vibrational dynamics of the active medium in CO_2 amplifiers. In this model, the following temperatures represent the distribution of energy among molecular vibrations:

- T_2 : Vibrational temperature of the ν_1 and ν_2 modes of CO_2 .
- T_3 : Vibrational temperature of the ν_3 mode of CO_2 .
- T_4 : Vibrational temperature of N_2 .

The temperature model adopted in `co2amp` relates the average number of quanta in each vibrational mode of the CO_2 molecule to a mode temperature, as described by the first two equations in 4.3. The temperatures of the ν_1 and ν_2 modes, coupled by Fermi resonance, are considered equal, and thus only T_2 is used to describe both modes.

Vibrational temperatures are related to the average number of quanta e_x in the corresponding vibrations as follows:

$$\begin{aligned}
e_1 &= \frac{1}{\exp\left(\frac{1920}{T_2}\right) - 1}, \\
e_2 &= \frac{2}{\exp\left(\frac{960}{T_2}\right) - 1}, \\
e_3 &= \frac{1}{\exp\left(\frac{3380}{T_3}\right) - 1}, \\
e_4 &= \frac{1}{\exp\left(\frac{3350}{T_4}\right) - 1}.
\end{aligned} \tag{4.3}$$

The factor of 2 in the second equation accounts for the two-fold degeneracy of the energy levels of the bending vibration.

In thermal equilibrium, the vibrational energy distribution follows the Boltzmann distribution. For the groups of vibrational levels defined in Table 4.1, this distribution is expressed as [5]:

$$\begin{aligned}
N_{000} &= \frac{N}{\mathcal{Q}}, \\
N_{001} &= \frac{N}{\mathcal{Q}} \exp\left(-\frac{3380}{T_3}\right), \\
N_{002} &= \frac{N}{\mathcal{Q}} \exp\left(-\frac{2 \times 3380}{T_3}\right), \\
N_{003} &= \frac{N}{\mathcal{Q}} \exp\left(-\frac{3 \times 3380}{T_3}\right), \\
N_{100+020} &= 4 \frac{N}{\mathcal{Q}} \exp\left(-\frac{2 \times 960}{T_2}\right), \\
N_{110+030} &= 6 \frac{N}{\mathcal{Q}} \exp\left(-\frac{3 \times 960}{T_2}\right), \\
N_{011} &= 2 \frac{N}{\mathcal{Q}} \exp\left(-\frac{960}{T_2}\right) \exp\left(-\frac{3380}{T_3}\right), \\
N_{101+021} &= 4 \frac{N}{\mathcal{Q}} \exp\left(-\frac{2 \times 960}{T_2}\right) \exp\left(-\frac{3380}{T_3}\right), \\
N_{201+121+041} &= 9 \frac{N}{\mathcal{Q}} \exp\left(-\frac{4 \times 960}{T_2}\right) \exp\left(-\frac{3380}{T_3}\right).
\end{aligned} \tag{4.4}$$

Here, N is the density of CO_2 molecules, and $\mathcal{Q}$ is the partition function [5, 6]:

$$\frac{1}{\mathcal{Q}} = \left(1 - \exp\left(-\frac{1920}{T_2}\right)\right) \cdot \left(1 - \exp\left(-\frac{3380}{T_3}\right)\right) \cdot \left(1 - \exp\left(-\frac{960}{T_2}\right)\right)^2. \tag{4.5}$$

After calculating the total equilibrium population of each group of vibrational levels listed in Table 4.1, the equilibrium population of the individual levels within a group is obtained by distributing the group population according to Boltzmann factors based on their relative energies. For groups in which all levels contain the same number of ν_3 quanta (fixed v_3), the common ν_3 contribution cancels in the normalization, so only the residual energy splittings (Fermi and ℓ -doubling) are relevant for the intra-group distribution.

Coefficients in Eq. 4.4 correspond to the degeneracy (number of vibrational levels) in each group (see Table 4.1).

The intra-mode vibrational energy thermalization process is typically very fast. However, if the partial pressure of CO_2 is very low, or if the active medium is pumped by a very short optical pulse, the finite

thermalization time must be considered. The thermalization time, τ_V , is assumed to be the same for all vibrational modes and is adopted from [7], where the relaxation time of the (10^01) and (02^01) levels is given as:

$$\tau_V [\text{s}] = \frac{10^{-6}}{750 (3.9 P_{\text{CO}_2})}, \quad (4.6)$$

where the pressure P_{CO_2} is measured in bar. Note that since intra-mode vibrational energy thermalization only involves collisional energy transfer between CO_2 molecules, the thermalization time does not depend on the partial pressures of other gases, such as He and N_2 .

During pumping and amplification, the populations of vibrational levels are treated independently for each CO_2 isotopologue. Thermalization is then accounted for on the "slow" lab time frame. Due to the lack of data on inter-isotopologue vibrational energy transfer rates and to simplify calculations, the same relaxation time, τ_V , is used for both inter-mode and intra-isotopologue thermalization.

It is assumed that the vibrational temperatures T_2 and T_3 are identical for all CO_2 isotopologues. This assumption is justified by the small energy mismatch between vibrational levels of different isotopic species, allowing rapid inter-molecular V-V energy exchange.

In the **co2amp** model, pumping and vibrational relaxation processes are typically slow compared to the duration of the laser pulse. Therefore, only stimulated transitions are taken into account for population changes during pulse amplification. To account for rare scenarios involving relatively fast vibrational relaxation (e.g., relatively high CO_2 concentrations and a relatively long amplified pulse), the amplification can be modeled in several steps, with vibrational dynamics calculations performed in between. This multi-step approach is engaged automatically when the time step of the *lab time frame* is shorter than the overall duration of the *pulse time frame*.

Rotational sub-level populations are also subject to thermalization. However, the corresponding thermalization time, τ_R , depends on collisions with all components of the active medium, not just CO_2 . This time can be fast and comparable to the pulse duration, so τ_R is included in the pulse time frame calculations modeling pulse amplification, as described Section 4.3

4.2.3 Pumping by electric discharge

Simulations of active medium pumping by electric discharge are conducted following the approach described by Karlov and Konev [8].

Pumping is modeled using the Boltzmann equation in the following form [9, 10]:

$$\begin{aligned} -\frac{1}{3} \left(\frac{\mathcal{E}}{\mathcal{N}} \right)^2 \frac{d}{du} \left[u \left(\sum_j y_j Q_{mj}(u) \right)^{-1} \frac{df(u)}{du} \right] = \\ 1.09 \times 10^{-3} \frac{d}{du} \left[u^2 f(u) \sum_j \frac{y_j}{M_j} Q_{mj}(u) \right] + \sum_{j=1,2} y_j C_j \frac{d}{du} (u f(u)) + 6 B y_2 \frac{d}{du} (u Q(u) f) \\ + \sum_j y_j \sum_k (u + u_{jk}) Q_{jk}(u + u_{jk}) f(u + u_{jk}) - u f(u) \sum_j y_j \sum_k Q_{jk}(u) \end{aligned} \quad (4.7)$$

where the terms represent the following:

- The left-hand side describes the energy of electrons in the electric field.
- The first term on the right-hand side represents energy transfer via elastic collisions between electrons and molecules.
- The second and third terms describe collisions involving rotational excitation of molecules.

- The last two terms account for inelastic collisions that transfer energy u_{jk} into vibrational and electronic excitations, as well as ionization.

Here:

- Electron energy u is expressed in electronvolts (eV).
- The ratio of the electric field to the full molecular density, $\mathcal{E}/\mathcal{N}$, is expressed in units of 10^{-16} V cm^2 .
- y_j are the relative molecule concentrations ($j = 1$ corresponds to CO_2 , $j = 2$ to N_2 and $j = 3$ to He);
- $M_1 = 44$, $M_2 = 28$, $M_3 = 4$ are the molar masses;
- $C_1 = 8.2 \times 10^{-4} \text{ eV \AA}^2$ [11];
- $C_2 = 5.06 \times 10^{-4} \text{ eV \AA}^2$ [12];
- $B = 2.5 \times 10^{-4} \text{ eV}$ is the N_2 rotational constant.
- Numerical values of the cross-sections Q and the transferred energies u_{jk} are summarized in Appendix A

Equation 4.7 is solved numerically using the tridiagonal matrix algorithm. Distribution function $f(u)$ is then used in the following calculations.

The rate constant ω_{jk} , and the electron drift speeds v_d are defined as:

$$\omega_{jk} \left[\frac{\text{cm}^3}{\text{s}} \right] = 5.93 \times 10^{-9} \int_0^\infty u Q_{jk}(u) f(u) du \quad (4.8)$$

$$v_d \left[\frac{\text{cm}}{\text{s}} \right] = -5.93 \times 10^7 \left(\frac{1}{3} \frac{\mathcal{E}}{\mathcal{N}} \right) \frac{df(u)}{du} \int_0^\infty u \left(\sum_j y_j Q_{mj}(u) \right)^{-1} du \quad (4.9)$$

The fraction of electron energy transmitted via inelastic processes is defined as

$$z_{jk} = 10^{16} \frac{y_j u_{jk} \omega_{jk}}{\left(\frac{\mathcal{E}}{\mathcal{N}} \right) v_d} \quad (4.10)$$

The fraction of electron energy transmitted to translations and rotations are the following:

$$z_t = 5.93 \times 10^7 \frac{1.09 \times 10^{-3} \int_0^\infty u^2 \left(\sum_j \frac{y_j}{M_j} Q_{mj}(u) \right) f(u) du}{\left(\frac{\mathcal{E}}{\mathcal{N}} \right) v_d} \quad (4.11)$$

$$z_r = 5.93 \times 10^7 \frac{\sum_{j=1,2} y_j C_j \int_0^\infty u f(u) du + 6 y_2 B \int_0^\infty u Q(u) f(u) du}{\left(\frac{\mathcal{E}}{\mathcal{N}} \right) v_d} \quad (4.12)$$

The distribution of the excitation energy is calculated using the following expressions:

- $q_2 = \sum_{k=1}^6 z_{1k}$ - fraction of energy transferred to CO_2 symmetric stretch (ν_1) and bending (ν_2) modes;
- $q_3 = z_{17}$ - fraction of energy transferred to CO_2 asymmetric stretch mode (ν_3);
- $q_4 = \sum_{k=1}^8 z_{2k}$ - fraction of energy transferred to N_2 vibrations;
- $q_T = z_t + z_r$ - fraction of energy transferred to translation and rotation;

$q_{ei} = \sum_{k=9}^{15} z_{2k} + \sum_{k=8}^{10} z_{1k}$ - fraction of energy spent on excitation of electronic levels and ionization.

Finally, we can calculate the pumping rates for each of the vibrational modes considered in the model:

$$\begin{aligned} p_{e4} &= 0.8 \times 10^{-3} \frac{q_4}{ny_2} W(t), \\ p_{e3} &= 0.8 \times 10^{-3} \frac{q_3}{ny_1} W(t), \\ p_{e2} &= 2.8 \times 10^{-3} \frac{q_2}{ny_1} W(t), \end{aligned} \quad (4.13)$$

where $W(t)$ is the discharge power density measured in kW/cm^3 and $n = 273 \frac{p[\text{bar}]}{T_0[\text{K}]}$ is the total concentration of gas molecules in the active medium relative to concentration at 1 bar and 273 K.

4.2.4 Optical Pumping

For optical pumping, the pumping rates are calculated using the following expressions:

$$\begin{aligned} p_{e4} &= 0, \\ p_{e3} &= \begin{cases} \Phi \sigma \frac{N_{gs} - N^*}{N_{CO_2}} & \text{direct excitation of } (00^01) \text{ at } \sim 4.3 \mu\text{m}, \\ \Phi \sigma \frac{N_{gs} - \tilde{N}^*}{N_{CO_2}} & \text{excitation via } (10^01, 02^01) \text{ at } \sim 2.8 \mu\text{m}, \\ \Phi \sigma \frac{N_{gs} - \tilde{N}^*}{N_{CO_2}} & \text{excitation via } (20^01, 12^01, 04^01) \text{ at } \sim 2.0 \mu\text{m}, \\ 2\Phi \sigma \frac{N_{gs} - N^*}{N_{CO_2}} & \text{excitation via } (00^02) \text{ at } \sim 2.2 \mu\text{m}, \\ 3\Phi \sigma \frac{N_{gs} - \tilde{N}^*}{N_{CO_2}} & \text{excitation via } (00^03) \text{ at } \sim 1.4 \mu\text{m}. \end{cases} \\ p_{e2} &= \begin{cases} 0 & \text{direct excitation of } (00^01) \text{ at } \sim 4.3 \mu\text{m}, \\ 2\Phi \sigma \frac{N_{gs} - N^*}{N_{CO_2}} & \text{excitation via } (10^01, 02^01) \text{ at } \sim 2.8 \mu\text{m}, \\ 4\Phi \sigma \frac{N_{gs} - \tilde{N}^*}{N_{CO_2}} & \text{excitation via } (20^01, 12^01, 04^01) \text{ at } \sim 2.0 \mu\text{m}, \\ 0 & \text{excitation via } (00^02) \text{ at } \sim 2.2 \mu\text{m}, \\ 0 & \text{excitation via } (00^03) \text{ at } \sim 1.4 \mu\text{m}. \end{cases} \end{aligned} \quad (4.14)$$

Here, Φ is the flux of the pumping photons (number of photons per m^2 per second), σ is the absorption cross-section, N_{gs} and N^* are the densities of CO_2 molecules on the ground state and the excitation level, and N_{CO_2} is the total density of CO_2 molecules.

Equations 4.14 define the number of vibrational quanta delivered by each pump photon to the modes associated with the upper and lower laser levels. The corresponding quanta are added directly to the populations of the relevant levels. This makes it possible to model amplification accurately before thermal equilibrium is established among the vibrational modes. Such a non-equilibrium treatment is particularly important for amplification on the $\sim 4 \mu\text{m}$ transitions originating from the short-lived $(10^01, 02^01)$ and 02^21 levels.

4.2.5 Pumping and vibrational relaxation dynamics

The dynamics of pumping and vibrational relaxation are modeled in the lab time frame and are described by the following equations:

$$\begin{aligned} \frac{de_4}{dt} &= p_{e4} - r_a(e_4 - e_3), \\ \frac{de_3}{dt} &= p_{e3} + r_c(e_4 - e_3) - r_3 f_3, \\ \frac{de_2}{dt} &= f_2(p_{e2} + 3r_3 f_3 - r_2(e_2 - e_{2T})), \end{aligned} \quad (4.15)$$

where

$$\begin{aligned}
f_2 &= \frac{2(1+e_2)^2}{2+6e_2+3e_2^2}; \quad f_3 = e_3(1+e_2/2)^3 - (1+e_3)(e_2/2)^3 \exp(-500/T); \\
r_a &= kny_1; \quad r_c = kny_2; \quad r_2 = k_2n; \quad r_3 = k_3n; \\
k_2 &= \sum_{i=1}^3 y_i k_{2i}; \quad k_3 = \sum_{i=1}^3 y_i k_{3i}; \\
e_{2T} &= \frac{2}{\exp(960/T) - 1}
\end{aligned} \tag{4.16}$$

Here, p_e is measured in μs^{-1} , and the constants k are calculated using the following expressions [13, 14]:

$$\begin{aligned}
k &= 240/T^{1/2}; \\
k_{31} &= A(t) \exp(4.138 + 7.945x - 631.24x^2 + 2239x^3); \\
k_{32} &= A(t) \exp(-1.863 + 213.3x - 2796.2x^2 + 9001.9x^3); \\
k_{33} &= A(t) \exp(-3.276 + 291.4x - 3831.8x^2 + 12688x^3); \\
k_{21} &= 1.16 \times 10^3 \exp(-59.3x); \\
k_{22} &= 8.55 \times 10^2 \exp(-69x); \\
k_{23} &= 1.3 \times 10^3 \exp(-40.6x)
\end{aligned} \tag{4.17}$$

where $x = T^{-1/3}$, $A(t) = (T/273)(1 + e_{2T}/2)^{-3}$, and the temperature T is expressed in kelvins (K). Finally, the dynamics of the gas temperature are described by the following equation:

$$\frac{dT}{dt} = \frac{y_1}{C_V} (500r_3f_3 + 960r_2(e_2 - e_{2T})) + 2.7 \frac{W(t)q_T}{nC_V}, \tag{4.18}$$

where $C_V = 2.5(y_1 + y_2) + 1.5y_3$.

4.3 Amplification

Amplification is simulated in the pulse time frame using the following equations, which also account for rotational relaxation [15, 16]:

$$\begin{aligned}
\frac{\partial E}{\partial z} &= - \sum_j \rho_j, \\
\frac{\partial \rho_j}{\partial t} + \left(2\pi i(\nu_c - \nu_j) + \frac{1}{\tau_2} \right) \rho_j &= - \frac{\sigma_j n_j E}{2\tau_2}, \\
\frac{\partial n_j}{\partial t} + \frac{n_j - n_j^0}{\tau_R} &= 4(\rho_j E^* + c.c.),
\end{aligned} \tag{4.19}$$

where the summation is performed over all rotational-vibrational transitions of all CO_2 isotopologues, and

- E — complex field envelope,
- ρ_j — polarization of the medium,
- z — coordinate along the direction of beam propagation,
- t — time,
- n_j — population inversion of transition j (difference of population densities of the upper and lower levels),
- n_j^0 — equilibrium population inversion of transition j ,
- ν_c — carrier frequency,
- ν_j — line-center frequency of transition j ,
- σ_j — line-center cross-section of transition j ,

τ_2 — *effective* polarization dephasing time used in the model,
 τ_R — rotational relaxation time.

Line-center cross-section and effective linewidth

The line-center cross-section is calculated using the expression [17]:

$$\sigma_j [\text{m}^2] = \frac{(\lambda_j [\text{m}])^2 A_j [\text{s}^{-1}]}{4} \times \frac{1}{\pi \Gamma [\text{rad/s}]} . \quad (4.20)$$

Here, the first factor corresponds to the integrated cross-section of the transition, while the second factor is the peak value of a normalized Lorentzian line shape expressed in angular frequency,

$$g(\omega) = \frac{1}{\pi} \frac{\Gamma}{(\omega - \omega_j)^2 + \Gamma^2}, \quad \int_{-\infty}^{\infty} g(\omega) d\omega = 1, \quad (4.21)$$

for which $g(\omega_j) = 1/(\pi\Gamma)$. The parameter Γ is the half-width at half-maximum (HWHM) in units of rad/s. When using a full-width at half-maximum (FWHM) in frequency units, $\Delta\nu$ in Hz, the conversion is

$$\Gamma [\text{rad/s}] = \pi \Delta\nu [\text{Hz}]. \quad (4.22)$$

In the model, the effective linewidth $\Delta\nu$ is approximated by a quadratic sum of a Lorentzian (collisional) contribution $\Delta\nu_L$ and a Doppler contribution $\Delta\nu_D$,

$$\Delta\nu = \sqrt{\Delta\nu_L^2 + \Delta\nu_D^2}, \quad \Delta\nu_L = \frac{1}{\pi \tau_{2,L}}, \quad \Delta\nu_D = \nu_j \sqrt{\frac{8k_B T_0 \ln 2}{mc^2}}, \quad (4.23)$$

where $\tau_{2,L}$ is the *real* (collisional) coherence (dephasing) time associated with the Lorentzian homogeneous broadening, ν_j is the line-center frequency of transition j , k_B is the Boltzmann constant, T_0 is the gas temperature, m is the mass of a CO_2 molecule, and c is the speed of light. This quadratic combination is used to maintain internal consistency with the Lorentzian line-shape treatment in the **co2amp** amplification model, with Doppler broadening included as a correction rather than a full Voigt-profile treatment.

For consistency with Eq. (4.19) (which assumes a Lorentzian dephasing term $1/\tau_2$), **co2amp** uses an *effective* dephasing time τ_2 defined by the effective linewidth:

$$\tau_2 \equiv \tau_{2,\text{eff}} = \frac{1}{\pi \Delta\nu} = \frac{1}{\pi \sqrt{\Delta\nu_L^2 + \Delta\nu_D^2}}. \quad (4.24)$$

With this definition, Eq. (4.19) can be used without modification, while Doppler broadening is approximately accounted for via $\tau_{2,\text{eff}}$.

Small-signal gain in this model is obtained by summing the contributions from all transitions,

$$G(\nu) [\text{m}^{-1}] = \sum_j \sigma_j [\text{m}^2] \Delta N_j [\text{m}^{-3}] \frac{(\Delta\nu_j/2)^2}{(\nu - \nu_j)^2 + (\Delta\nu_j/2)^2}, \quad (4.25)$$

where ν_j is the line-center frequency of transition j , ΔN_j is the population inversion density for that transition, and $\Delta\nu_j$ is the Lorentzian full-width at half-maximum (FWHM) expressed in Hz. With this convention, the Lorentzian factor equals unity at line center, so that σ_j is the line-center cross-section.

Optical intensity is related to the complex field amplitude used in the model via

$$I [\text{W/m}^2] = 2 h [\text{J s}] \nu_c [\text{s}^{-1}] |E|^2 [\text{m}^{-2} \text{s}^{-1}]. \quad (4.26)$$

With this normalization, $2|E|^2$ corresponds to the photon flux density (photons per unit area per unit time), i.e., the optical energy flux I divided by the photon energy $h\nu_c$. The prefactor 2 is a convention of the complex-amplitude normalization used in the model.

Dephasing and relaxation times

The collisional dephasing time $\tau_{2,L}$ and rotational relaxation time τ_R are defined as

$$\begin{aligned}\tau_{2,L} [\text{s}] &= \frac{10^{-6}}{\pi \times 7.61 \times 750 \times (P_{\text{CO}_2} + 0.73 P_{\text{N}_2} + 0.64 P_{\text{He}})}, \\ \tau_R [\text{s}] &= \frac{10^{-7}}{750 \times (1.3 P_{\text{CO}_2} + 1.2 P_{\text{N}_2} + 0.6 P_{\text{He}})},\end{aligned}\tag{4.27}$$

where the partial pressures P are measured in bar. The effective dephasing time $\tau_2 = \tau_{2,\text{eff}}$ used in Eq. (4.19) is obtained from $\tau_{2,L}$ via Eqs. (4.23)–(4.24).

The population change due to stimulated transitions is calculated for each vibrational level of each isotopologue using the last equation in Eq. (4.19):

$$\begin{aligned}\frac{d}{dt}N_U &= 2 \sum_j (\rho_j E^* + \text{c.c.}), \\ \frac{d}{dt}N_L &= -2 \sum_j (\rho_j E^* + \text{c.c.}),\end{aligned}\tag{4.28}$$

where ρ_j is the coherence for transition j , c.c. denotes the complex conjugate, and the summation is over all rotational transitions originating from (N_U) or terminating at (N_L) the corresponding vibrational level.

Next, we compute the net changes in the ν_2 and ν_3 quanta, denoted ΔN_{ν_2} and ΔN_{ν_3} , by summing the population changes over all vibrational levels and all isotopologues, weighting each level by the number of corresponding quanta. In this accounting, each ν_1 quantum is treated as equivalent to two ν_2 quanta, consistent with the assumption that the ν_1 and ν_2 modes share a common vibrational temperature ($T_1 \approx T_2$) due to Fermi coupling.

The changes in the average vibrational quantum numbers due to stimulated transitions are then

$$\begin{aligned}\Delta e_3 &= \frac{\Delta N_{\nu_3}}{N}, \\ \Delta e_2 &= \frac{\Delta N_{\nu_2}}{N} \times \frac{e'_2}{2e'_1 + e'_2},\end{aligned}\tag{4.29}$$

where N is the total molecular number density. The factor in the second line accounts for the equilibrium energy partition between the coupled symmetric stretch (ν_1) and bending (ν_2) modes. Here, e'_1 and e'_2 are the equilibrium average quantum numbers,

$$e'_1 = \frac{1}{\exp\left(\frac{1920}{T_2}\right) - 1}, \quad e'_2 = \frac{2}{\exp\left(\frac{960}{T_2}\right) - 1},\tag{4.30}$$

with T_2 the vibrational temperature (in K) before pulse propagation. New vibrational temperatures are then obtained using Eq. (4.3).

Appendices

Appendix A

Cross-sections of excitation processes

Effective cross-sections are expressed in Å; their numerical values in the nodes are given in the tables below (linear interpolation must be used for determining the values in intermediate points); the data and citations are reproduced from [8].

The following notation for cross-sections is used:

Q_{m1} - Transport cross-section of CO₂ [18];

Q_{m2} - Transport cross-section of N₂ [12];

Q_{m3} - Transport cross-section of He [18];

Q - Cross-section of resonant excitation of N₂ rotation [19, 20];

Q_{11} - Cross-section of the process $(000) \rightarrow (01^10)$ [18];

Q_{12} - Cross-section of the process $(000) \rightarrow (100 + 020)$ [18];

$Q_{13}...Q_{16}$ - Cross-sections of resonant processes around 3.8 eV [18];

Q_{17} - Cross-section of the process $(000) \rightarrow (001)$ [18];

$Q_{18}...Q_{1,10}$ - Cross-sections of electronic excitation and ionization of CO₂ [11];

$Q_{21}...Q_{28}$ - Cross-sections of the process $N_2(v=0) \rightarrow N_2(v=1...8)$ [21, 22, 23];

$Q_{29}...Q_{2,15}$ - Cross-sections of electronic excitation and ionization of N₂ [23].

Table A.1: Cross-sections and energies for discharge pumping

u_i	Q_{m1}	u_i	Q_{m2}	u_i	Q_{m3}	u_i	Q
0	140	0	1.4	0	5	0.0015	0
0.04	84	0.001	1.4	0.01	5.4	0.05	0.1
0.1	55	0.002	1.6	0.1	5.8	0.25	0.65
0.3	21	0.008	2	0.2	6.2	0.5	1.15
0.5	10.8	0.01	2.2	1	6.5	0.8	2
0.6	9.4	0.04	4	2	6.1	1	2.65
1	5.7	0.08	6	7	5	1.5	5.6
1.7	5	0.1	6.5	10	4.1	1.8	7.5
2	5.1	0.2	8.8	20	3	1.9	8.2
2.5	6	0.3	9.8			2	8.6
3	7.7	0.4	10			2.15	8.95
4.1	9.4	1	10			2.43	9
5	14.5	1.2	11			2.6	8.9
7.4	10	1.4	12.5			2.75	8.4
10	11.7	1.8	20			2.9	7.65
20	16	2	25			3.25	6.2
27	16.3	2.5	30			3.6	5.1
50	13	3	26			4	4.5
		4	15			4.5	4.16
		5	12			5	3.97
		7	10			5.5	3.93
		10	10			7	4.17
		14	11			9	4.46
		18	12.2			11	4.42
		20	12			15	3.94
		30	10			22	3.15
		100	10			25	3.05

Table A.2: Cross-sections and energies for discharge pumping - continued

u_i	Q_{11}	u_i	Q_{12}	u_i	Q_{13}	u_i	Q_{14}	u_i	Q_{15}
0.083	0	0.167	0	0.252	0	2.37	0	2.37	0
0.085	0.36	0.2	0.54	2.7	0.25	3	0.26	3	0.17
0.09	1.04	0.25	0.82	3	0.4	3.5	0.52	3.65	0.33
0.1	1.6	0.3	0.82	3.3	0.6	4	0.5	3.8	0.31
0.12	1.84	0.5	0.68	3.6	0.65	4.5	0.22	4	0.21
0.14	2.12	0.7	0.56	4.5	0.23	4.6	0.1	4.3	0.1
0.16	2.16	1	0.47	4.6	0.1	5	0	5	0
0.2	2.08	1.4	0.45	5	0				
0.3	1.76	2	0.55						
0.4	1.52	3	1.15						
0.5	1.28	3.9	1.83						
0.6	1.08	4.5	1.4						
0.8	0.8	5	0.4						
1	0.58	6	0.28						
1.2	0.48	10	0.2						
1.6	0.34	20	0.1						
1.8	0.35								
2	0.4								
2.5	0.64								
3	1.04								
3.7	1.4								
4	1.36								
4.2	1.2								
4.5	0.92								
5	0.53								
6	0.4								
8	0.36								
9	0.28								
10	0.16								
10.1	0								
$u_{11} = 0.083 \text{ eV}$		$u_{12} = 0.167 \text{ eV}$		$u_{13} = 0.252 \text{ eV}$		$u_{14} = 0.339 \text{ eV}$		$u_{15} = 0.422 \text{ eV}$	
u_i	Q_{16}	u_i	Q_{17}	u_i	Q_{18}	u_i	Q_{19}	u_i	$Q_{1,10}$
2.5	0	0.29	0	7	0	10.5	0	13.8	0
3	0.19	0.3	0.44	8	0.5	11.5	0.56	15	0.1
3.6	0.245	0.35	0.65	8.4	0.6	14	0.8	16	0.13
4	0.21	0.4	0.73	9	0.46	20	1.2	17	0.17
5.07	0	0.5	0.84	10	0.175	30	2	30	1.55
		0.8	1	10.5	0	50	4	40	2.1
		1	1						
		2	0.78						
		6	0.37						
		10	0.25						
		50	0						
$u_{16} = 2.5 \text{ eV}$		$u_{17} = 0.29 \text{ eV}$		$u_{18} = 7 \text{ eV}$		$u_{19} = 10.5 \text{ eV}$		$u_{1,10} = 13.8 \text{ eV}$	

Table A.3: Cross-sections and energies for discharge pumping - continued

u_i	Q_{21}	u_i	Q_{22}	u_i	Q_{23}	u_i	Q_{24}	u_i	Q_{25}
0.29	0	1.83	0	1.9	0	2.05	0	2.1	0
0.5	0.0052	1.9	0.208	2	0.416	2.1	0.416	2.15	0.208
0.8	0.0083	2	1.46	2.1	1.33	2.2	1.16	2.2	0.541
1	0.0104	2.05	2.29	2.2	1.87	2.26	1.58	2.3	0.915
1.2	0.0166	2.1	1.66	2.3	1.25	2.55	0	2.46	1.12
1.3	0.0728	2.2	0.79	2.36	0.208	2.75	0.832	2.5	1.12
1.4	0.135	2.35	0.208	2.42	0	2.77	0	2.6	0.208
1.6	0.25	2.45	1.98	2.5	0.499	3	0.208	2.62	0
1.8	0.52	2.5	1.78	2.61	0.915	3.05	0.208	2.68	0
1.9	0.832	2.62	0.208	2.7	0.624	3.25	0	2.8	0.416
2	3.02	2.75	1.04	2.75	0.208			2.9	0.75
2.05	3.12	2.95	1.66	2.8	0			3	0
2.1	2.08	3.05	0.624	2.92	0.416			3.2	0.25
2.15	1.25	3.2	0.208	3	0.208			3.3	0.125
2.2	0.832	3.4	0.208	3.25	0.208			3.35	0
2.3	2.9	4	0	3.31	0				
2.45	1.04								
2.53	1.25								
2.6	1.75								
2.62	2.08								
2.68	1.73								
2.73	0.416								
2.85	0.32								
2.92	0.416								
3.12	0.728								
3.3	0.52								
4	0								
$u_{21} = 0.29 \text{ eV}$		$u_{22} = 0.58 \text{ eV}$		$u_{23} = 0.87 \text{ eV}$		$u_{24} = 1.16 \text{ eV}$		$u_{25} = 1.45 \text{ eV}$	
u_i	Q_{26}	u_i	Q_{27}	u_i	Q_{28}	u_i	Q_{29}	u_i	$Q_{2,10}$
2.3	0	2.4	0	2.6	0	5	0	6.8	0
2.4	0.75	2.5	0.208	2.7	0.208	5.9	0.41	7.1	0.57
2.5	1.04	2.75	0.75	2.9	0.29	6.1	0.41	8.1	0.57
2.55	1.12	3	0	3	0.208	7	0.07	8.6	0.25
2.6	1.04	3.2	0.166	3.1	0	9	0	9.5	0.12
2.65	0.624	3.3	0.146	3.2	0			20.7	0
2.7	0.416	3.4	0	3.3	1.04				
2.8	0.208			3.4	0				
2.9	0.125								
3	2.5								
3.1	0.166								
3.2	0								
$u_{26} = 1.74 \text{ eV}$		$u_{27} = 2.03 \text{ eV}$		$u_{28} = 2.32 \text{ eV}$		$u_{29} = 5 \text{ eV}$		$u_{2,10} = 6.8 \text{ eV}$	
u_i	$Q_{2,11}$	u_i	$Q_{2,12}$	u_i	$Q_{2,13}$	u_i	$Q_{2,14}$	u_i	$Q_{2,15}$
8.4	0	11.25	0	12.5	0	14	0	15.6	0
8.7	0.42	13.8	0.41	13	0.4	14.3	1.7	18	0.1
9.1	0.42	14	1	13.6	0.4	14.8	1.7	20	0.21
10	0.3	14.7	1	14	0.16	15.6	0.2	50	2.52
20.7	0	15	0.25	20.7	0	20.6	0.2	100	2.52
		65	0			25.4	2.8		
						100	2.8		
$u_{2,11} = 8.4 \text{ eV}$		$u_{2,12} = 11.25 \text{ eV}$		$u_{2,13} = 12.5 \text{ eV}$		$u_{2,14} = 14 \text{ eV}$		$u_{2,15} = 15.6 \text{ eV}$	

Appendix B

Molecular constants

In most cases, the transition frequencies and Einstein coefficients (A) for every rotational line of supported ro-vibrational bands are extracted directly from a HITRAN database `.par` file. However, the current version of HITRAN (HITRAN2020 [4]) lacks information on certain transitions that could potentially make a significant contribution to the amplification process. To address these gaps, we have supplemented the most critical missing data with rough estimates to enable more realistic modeling, albeit with limited precision.

Table B.1 lists the number of rotational lines in the P- and R-branches of each ro-vibrational band included in the **co2amp** model. The numbering of the ro-vibrational bands is consistent with Table 4.2. Table B.2 provides the Einstein coefficients (A) averaged over all P- and R-branch rotational lines included in HITRAN for a given ro-vibrational band. Note that the model uses the actual A values for each line; the averaged values are presented here for reference only. The model includes only those lines where the rotational quantum numbers J for both the upper and lower levels do not exceed 79.

The model considers only P- and R-branch transitions. Although HITRAN includes Q-branch transitions for some bands in certain cases, their contribution to amplification is negligible due to their intensities being at least two orders of magnitude lower than those of the P- and R-branches.

Zeros in Table B.1 indicate that no rotational lines for a given ro-vibrational bands are present in HITRAN. Additionally, for some critical transitions, the number of included lines is very low (e.g., the first regular band for the 838 isotopologue has only 10 listed P- and R-branch lines combined). To address these critical gaps, we use the following approach:

1. Calculate the frequencies of missing transitions using the molecular constants obtained via the fitting procedure described later in this Appendix.
2. When adding additional lines to a band that already has some listed in HITRAN, assign the average Einstein coefficient (A) of the existing lines to the added transitions.
3. If no rotational lines for a given band are listed in HITRAN, roughly estimate the likely A coefficient by interpolating data from other isotopologues and ro-vibrational bands.

In cases where estimated lines are included in the **co2amp** distribution as separate `.par` files, an asterisk is placed in the corresponding field of Table B.1, and the estimated A coefficient is given in Table ?? in square brackets. Users can remove these additional `.par` files from the `hitran_data` folder to exclude them from calculations, or add new `.par` files at their discretion. Care should be taken to avoid duplication of rotational lines across multiple `.par` files.

Molecular constants for vibrational levels, which can be used to calculate transition frequencies of ro-vibrational lines, including those not listed in the HITRAN database, can often be obtained by fitting the frequencies of the listed transitions. We have developed a fitting procedure that enables us to determine molecular constants for nearly all vibrational levels of all CO₂ isotopologues included in the **co2amp** model.

Table B.1: Numbers of P- and R-branch rotational lines in the HITRAN2020 database. Transitions for which estimated constants of missing rotational lines are included in the model are labeled by asterisks.

Band #	626	727	828	636	737	838	627	628	728	637	638	738
Regular bands												
0	99	24	57	87	0*	10*	152	165	93	119	135	0*
1	103	68	65	87	0*	28*	159	175	110	119	139	0*
Hot bands												
2e	89	0	30*	75	0	0	124	140	0	75	101	0
2f	89	0	29*	76	0	0	123	139	0	75	101	0
3e	92	0	44	77	0	0	133	149	44	79	107	0
3f	92	0	44	76	0	0	133	149	44	79	107	0
Sequence bands												
4	66	0	0*	44	0	0	0	13*	0	0	0	0
5	68	0	0*	41	0	0	0	60*	0	0	0	0
4-μm bands												
6	113	129	83	103	68	65	189	201	142	159	173	109
7	115	134	85	103	78	67	191	203	147	163	177	115
8e	111	127	81	101	72	48	185	197	138	157	171	109
8f	112	126	82	101	73	48	185	198	149	157	171	109

Table B.2: Einstein coefficients A for laser transitions, in s^{-1} (averaged over all P- and R-branch lines). Estimated values are given in square brackets.

Band #	626	727	828	636	737	838	627	628	728	637	638	738
Regular bands												
0	0.173	0.146	0.105	0.195	[0.18]	0.151	0.160	0.137	0.125	0.188	0.169	[0.16]
1	0.232	0.292	0.353	0.115	[0.17]	0.215	0.263	0.298	0.327	0.137	0.165	[0.19]
Hot bands												
2e	0.162	-	0.119	0.167	-	-	0.154	0.140	-	0.162	0.150	-
2f	0.161	-	0.118	0.166	-	-	0.154	0.139	-	0.162	0.150	-
3e	0.223	-	0.315	0.127	-	-	0.245	0.271	0.294	0.144	0.164	-
3f	0.225	-	0.317	0.128	-	-	0.246	0.272	0.295	0.144	0.164	-
Sequence bands												
4	0.391	-	[0.2]	0.422	-	-	-	0.310	-	-	-	-
5	0.420	-	[0.7]	0.200	-	-	-	0.543	-	-	-	-
4-μm bands												
6	196	195	188	176	173	167	196	193	192	175	172	171
7	197	195	186	178	174	168	196	192	191	177	174	172
8e	196	191	186	176	170	162	194	191	189	174	171	167
8f	197	191	186	176	170	163	194	191	189	174	171	167

The fitting is performed using the `hitran_FitAll.py` script. The fitting formula is as follows:

$$\nu(v, J) = G_v + B_v J(J + 1) - D_v [J(J + 1)]^2 + H_v [J(J + 1)]^3 + L_v [J(J + 1)]^4 \quad (\text{B.1})$$

To calculate the frequencies of missing transitions, we use the complete set of constants in formula (B.1) with full precision. The molecular dynamics model, however, uses only the G and B constants with truncated precision. The values of these constants as used in the code are given in Tables B.3 and B.4. Note that we also include constants for levels not currently used in the model; in these cases, the Level # entry is left blank. These data are intended for future code development.

Table B.3: Values of the constant G used in the model, in 10^{14} Hz.

Level	#	626	727	828	636	737	838	627	628	728	637	638	738
Ground State													
00001e	0	0	0	0	0	0	0	0	0	0	0	0	0
$\nu_3, 2\nu_3, 3\nu_3$													
00011e	1	0.704	0.699	0.694	0.685	0.679	0.674	0.702	0.699	0.696	0.682	0.679	0.676
00021e	2	1.401	1.390	1.380	1.362	-	1.341	1.396	1.391	1.385	1.357	1.352	-
00031e	3	2.090	-	-	2.033	-	-	-	2.075	-	-	-	-
ν_2													
01101e		0.200	0.198	0.197	0.194	0.193	0.191	0.199	0.199	0.198	0.194	0.193	0.192
01101f		0.200	0.198	0.197	0.194	0.193	0.191	0.199	0.199	0.198	0.194	0.193	0.192
$2\nu_2$													
10001e	4	0.416	0.409	0.404	0.411	0.402	0.395	0.413	0.409	0.406	0.406	0.402	0.399
10002e	5	0.385	0.377	0.369	0.379	0.373	0.366	0.381	0.378	0.373	0.376	0.373	0.370
02201e	6	0.400	0.397	0.394	0.389	0.386	0.383	0.399	0.397	0.396	0.387	0.386	0.384
02201f	7	0.400	0.397	0.394	0.389	0.386	0.383	0.399	0.397	0.396	0.387	0.386	0.384
$3\nu_2$													
11101e	8	0.623	0.614	0.607	0.611	-	0.593	0.618	0.614	0.610	0.606	0.601	0.597
11101f	9	0.623	0.614	0.607	0.611	-	0.593	0.618	0.614	0.610	0.606	0.601	0.597
11102e	10	0.579	0.570	0.560	0.569	-	0.551	0.575	0.570	0.565	0.564	0.560	0.556
11102f	11	0.579	0.570	0.560	0.569	-	0.551	0.575	0.570	0.565	0.564	0.560	0.556
03301e	12	0.601	0.596	0.592	0.584	-	0.574	0.598	0.596	0.594	0.581	0.579	0.576
03301f	13	0.601	0.596	0.592	0.584	-	0.574	0.598	0.596	0.594	0.581	0.579	0.576
$4\nu_2$													
20001e		0.839	-	0.816	0.825	-	-	0.832	0.827	0.821	0.817	0.810	-
20002e		0.801	-	0.768	0.793	-	-	0.792	0.784	0.775	0.784	0.776	-
20003e		0.764	0.749	0.734	0.752	-	-	0.757	0.750	0.742	0.746	0.740	-
12201e		0.828	-	0.809	0.810	-	-	0.822	0.818	0.813	0.804	0.799	-
12201f		0.828	-	0.809	0.810	-	-	0.822	0.818	0.813	0.804	0.799	-
12202e		0.775	0.764	0.753	0.759	-	0.739	0.769	0.764	0.758	0.754	0.749	-
12202f		0.775	0.764	0.753	0.759	-	0.739	0.769	0.764	0.758	0.754	0.749	-
04401e		0.801	0.795	0.789	0.778	-	0.766	0.798	0.795	0.792	0.775	0.772	-
04401f		0.801	0.795	0.789	0.778	-	0.766	0.798	0.795	0.792	0.775	0.772	-
$\nu_2 + \nu_3$													
01111e	14	0.901	0.893	0.887	0.875	0.868	0.862	0.897	0.894	0.890	0.872	0.869	0.865
01111f	15	0.901	0.893	0.887	0.875	0.868	0.862	0.897	0.894	0.890	0.872	0.869	0.865
$2\nu_2 + \nu_3$													
10011e	16	1.114	1.101	1.091	1.089	1.075	1.063	1.107	1.102	1.096	1.082	1.076	1.069
10012e	17	1.083	1.070	1.057	1.058	1.046	1.034	1.077	1.071	1.063	1.052	1.046	1.040
02211e	18	1.097	1.088	1.081	1.066	1.058	1.050	1.093	1.089	1.085	1.062	1.058	1.054
02211f	19	1.097	1.088	1.081	1.066	1.058	1.050	1.093	1.089	1.085	1.062	1.058	1.054
$3\nu_2 + \nu_3$													
11111e		1.316	1.302	1.290	1.285	1.27	1.257	1.309	1.303	1.296	1.278	1.271	1.263
11111f		1.316	1.302	1.290	1.285	1.27	1.257	1.309	1.303	1.296	1.278	1.271	1.263
11112e		1.273	1.259	1.245	1.243	1.229	1.216	1.266	1.259	1.252	1.236	1.230	1.223
11112f		1.273	1.259	1.245	1.243	1.229	1.216	1.266	1.259	1.252	1.236	1.230	1.223
03311e		1.294	1.283	1.274	1.258	-	1.238	1.289	1.284	1.279	1.252	1.248	1.242
03311f		1.294	1.283	1.274	1.258	-	1.238	1.289	1.284	1.279	1.252	1.248	1.242
$4\nu_2 + \nu_3$													
20011e	20	1.529	1.511	1.496	1.496	-	1.458	1.520	1.512	1.503	1.486	1.476	1.467
20012e	21	1.492	1.469	1.449	1.465	-	1.421	1.481	1.470	1.459	1.454	1.443	1.432
20013e	22	1.455	1.435	1.416	1.423	-	1.388	1.445	1.436	1.426	1.415	1.407	1.397
12211e	23	1.517	1.502	1.488	1.481	-	1.449	1.510	1.503	1.495	1.472	1.465	-
12211f	24	1.517	1.502	1.488	1.481	-	1.449	1.510	1.503	1.495	1.472	1.465	-
12212e	25	1.465	1.449	1.434	1.430	-	1.400	1.457	1.450	1.441	1.423	1.415	-
12212f	26	1.465	1.449	1.434	1.430	-	1.400	1.457	1.450	1.441	1.423	1.415	-
04411e	27	1.490	-	1.468	1.449	-	-	1.484	1.479	1.473	1.443	1.437	-
04411f	28	1.490	-	1.468	1.449	-	-	1.484	1.479	1.473	1.443	1.437	-

Table B.4: Values of the constant B used in the model, in 10^{10} Hz.

Level	#	626	727	828	636	737	838	627	628	728	637	638	738
Ground Stae													
00001e	0	1.170	1.101	1.040	1.170	1.101	1.040	1.135	1.104	1.070	1.135	1.104	1.070
$\nu_3, 2\nu_3, 3\nu_3$													
00011e	1	1.161	1.092	1.032	1.161	1.093	1.032	1.126	1.095	1.062	1.126	1.095	1.062
00021e	2	1.151	1.084	1.023	1.152	-	1.024	1.117	1.086	1.053	1.118	1.087	-
00031e	3	1.142	-	-	1.143	-	-	-	1.078	-	-	-	-
ν_2													
01101e		1.171	1.102	1.041	1.171	1.102	1.041	1.136	1.105	1.071	1.136	1.105	1.071
01101f		1.173	1.104	1.042	1.173	1.104	1.042	1.138	1.107	1.073	1.138	1.107	1.073
$2\nu_2$													
10001e	4	1.170	1.102	1.041	1.168	1.100	1.040	1.136	1.105	1.071	1.134	1.103	1.070
10002e	5	1.171	1.101	1.039	1.172	1.102	1.040	1.135	1.104	1.069	1.137	1.105	1.071
02201e	6	1.174	1.105	1.044	1.174	1.105	1.044	1.139	1.108	1.074	1.139	1.108	1.074
02201f	7	1.174	1.105	1.044	1.174	1.105	1.044	1.139	1.108	1.074	1.139	1.108	1.074
$3\nu_2$													
11101e	8	1.170	1.102	1.042	1.169	-	1.041	1.136	1.105	1.072	1.135	1.104	1.071
11101f	9	1.173	1.105	1.044	1.172	-	1.043	1.139	1.108	1.074	1.137	1.107	1.073
11102e	10	1.171	1.102	1.040	1.172	-	1.041	1.136	1.105	1.071	1.137	1.105	1.071
11102f	11	1.174	1.104	1.042	1.175	-	1.043	1.139	1.107	1.073	1.140	1.108	1.074
03301e	12	1.176	1.107	1.046	1.176	-	1.045	1.141	1.110	1.076	1.141	1.110	1.076
03301f	13	1.176	1.107	1.046	1.176	-	1.045	1.141	1.110	1.076	1.141	1.110	1.076
$4\nu_2$													
20001e		1.171	-	1.043	1.168	-	-	1.137	1.106	1.073	1.135	1.104	-
20002e		1.168	-	1.039	1.168	-	-	1.133	1.102	1.069	1.133	1.102	-
20003e		1.173	1.102	1.039	1.174	-	-	1.137	1.105	1.070	1.139	1.107	-
12201e		1.174	-	1.045	1.173	-	-	1.139	1.108	1.075	1.138	1.107	-
12201f		1.174	-	1.045	1.173	-	-	1.139	1.108	1.075	1.138	1.107	-
12202e		1.175	1.106	1.043	1.176	-	1.044	1.140	1.108	1.074	1.140	1.109	-
12202f		1.175	1.106	1.043	1.176	-	1.044	1.140	1.108	1.074	1.140	1.109	-
04401e		1.178	1.109	1.048	1.178	-	1.047	1.143	1.112	1.078	1.143	1.112	-
04401f		1.178	1.109	1.048	1.178	-	1.047	1.143	1.112	1.078	1.143	1.112	-
$\nu_2 + \nu_3$													
01111e	14	1.162	1.094	1.033	1.162	1.094	1.033	1.127	1.096	1.063	1.128	1.097	1.063
01111f	15	1.164	1.095	1.034	1.164	1.095	1.035	1.129	1.098	1.064	1.129	1.098	1.065
$2\nu_2 + \nu_3$													
10011e	16	1.160	1.093	1.033	1.159	1.092	1.032	1.126	1.096	1.063	1.125	1.095	1.062
10012e	17	1.162	1.092	1.031	1.163	1.094	1.032	1.127	1.095	1.061	1.128	1.097	1.063
02211e	18	1.165	1.096	1.036	1.165	1.097	1.036	1.130	1.099	1.066	1.131	1.099	1.066
02211f	19	1.165	1.097	1.036	1.165	1.097	1.036	1.131	1.099	1.066	1.131	1.099	1.066
$3\nu_2 + \nu_3$													
11111e		1.161	1.094	1.034	1.161	1.093	1.033	1.127	1.096	1.063	1.126	1.096	1.063
11111f		1.164	1.096	1.036	1.163	1.095	1.035	1.130	1.099	1.066	1.129	1.098	1.065
11112e		1.163	1.093	1.032	1.163	1.094	1.033	1.128	1.096	1.063	1.129	1.097	1.064
11112f		1.165	1.096	1.034	1.166	1.097	1.035	1.130	1.099	1.065	1.132	1.100	1.066
03311e		1.167	1.099	1.038	1.167	-	1.038	1.133	1.101	1.068	1.133	1.101	1.068
03311f		1.167	1.099	1.038	1.167	-	1.038	1.133	1.102	1.068	1.133	1.101	1.068
$4\nu_2 + \nu_3$													
20011e	20	1.162	1.095	1.035	1.159	-	1.034	1.128	1.098	1.065	1.126	1.096	1.063
20012e	21	1.160	1.091	1.031	1.160	-	1.031	1.124	1.093	1.061	1.125	1.094	1.060
20013e	22	1.164	1.094	1.031	1.166	-	1.034	1.128	1.096	1.062	1.131	1.099	1.065
12211e	23	1.165	1.097	1.036	1.164	-	1.036	1.131	1.100	1.066	1.130	1.099	-
12211f	24	1.165	1.097	1.036	1.164	-	1.036	1.131	1.100	1.066	1.130	1.099	-
12212e	25	1.166	1.098	1.036	1.167	-	1.036	1.131	1.100	1.067	1.132	1.102	-
12212f	26	1.166	1.098	1.036	1.167	-	1.036	1.131	1.100	1.067	1.132	1.102	-
04411e	27	1.170	-	1.040	1.170	-	-	1.135	1.104	1.070	1.135	1.103	-
04411f	28	1.170	-	1.040	1.170	-	-	1.135	1.104	1.070	1.135	1.103	-

Appendix C

Properties of optical materials

The following expressions and values for linear (n_0) and nonlinear (n_2) refractive indexes and linear absorption (α_0) are used in the program (wavelength λ in the formulas must be expressed in μm):

Air

Refractive index n_0 is calculated using Mathar's model for $\lambda = 7.5\text{--}14\ \mu\text{m}$ [24]

$$n_2 = 3.0 \times 10^{-23} \text{ m}^2/\text{W at } 9.2\ \mu\text{m} \text{ [25]}$$

AgBr

$$n_0 = \sqrt{3.860 + \frac{0.8677\lambda^2}{\lambda^2 - 0.3211^2} + \frac{21.61\lambda^2}{\lambda^2 - 254.2^2}} \quad (\lambda = 0.495\text{--}12.67\ \mu\text{m}) \text{ [26]}$$

$$n_2 = 6.0 \times 10^{-19} \text{ m}^2/\text{W at } 9.2\ \mu\text{m} \text{ [26]}$$

AgCl

$$n_0 = \sqrt{4.00804 + \frac{0.079086}{\lambda^2 - 0.04584} - 0.00085111\lambda^2 - 0.00000019762\lambda^4} \quad (\lambda = 0.578\text{--}20.6\ \mu\text{m}) \text{ [27]}$$

$$n_2 = 4.8 \times 10^{-19} \text{ m}^2/\text{W at } 9.2\ \mu\text{m} \text{ [26]}$$

BaF₂

$$n_0 = \sqrt{1.33973 + \frac{0.81070\lambda^2}{\lambda^2 - 0.10065^2} + \frac{0.19652\lambda^2}{\lambda^2 - 29.87^2} + \frac{4.52469\lambda^2}{\lambda^2 - 53.82^2}} \quad (\lambda = 0.15\text{--}15\ \mu\text{m}) \text{ [28]}$$

$$n_2 = 1.7 \times 10^{-20} \text{ m}^2/\text{W at } 9.2\ \mu\text{m} \text{ [29]}$$

$$\alpha_0 = 0.8(e^{1.20(\lambda-8)} - 1)\text{m}^{-1} \text{ [26]}$$

CdTe

$$n_0 = \sqrt{1 + \frac{6.1977889\lambda^2}{\lambda^2 - 0.1005326} + \frac{3.2243821\lambda^2}{\lambda^2 - 5279.518}} \quad (\lambda = 6\text{--}22\ \mu\text{m}) \text{ [30]}$$

$$n_2 = -2.95 \times 10^{-17} \text{ m}^2/\text{W at } 1.06\ \mu\text{m} \text{ [31]}$$

CsI

$$n_0 = \sqrt{1.27587 + \frac{0.68689\lambda^2}{\lambda^2-0.130^2} + \frac{0.26090\lambda^2}{\lambda^2-0.147^2} + \frac{0.06256\lambda^2}{\lambda^2-0.163^2} + \frac{0.06527\lambda^2}{\lambda^2-0.177^2} + \frac{0.14991\lambda^2}{\lambda^2-0.185^2} + \frac{0.51818\lambda^2}{\lambda^2-0.206^2} + \frac{0.01918\lambda^2}{\lambda^2-0.218^2} + \frac{3.38229\lambda^2}{\lambda^2-161.29^2}} \quad (\lambda = 0.25\text{--}67 \mu\text{m}) \quad [32]$$

$$n_2 = 1.2 \times 10^{-19} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [26]$$

GaAs

$$n_0 = \sqrt{5.372514 + \frac{5.466742\lambda^2}{\lambda^2-0.4431307^2} + \frac{0.02429960\lambda^2}{\lambda^2-0.8746453^2} + \frac{1.957522\lambda^2}{\lambda^2-36.9166^2}} \quad (\lambda = 0.97\text{--}17 \mu\text{m}) \quad [33]$$

$$n_2 = 7.5 \times 10^{-18} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [26]$$

Ge

$$n_0 = \sqrt{1 + \frac{0.4886331\lambda^2}{\lambda^2-1.393959} + \frac{14.5142535\lambda^2}{\lambda^2-0.1626427} + \frac{0.0091224\lambda^2}{\lambda^2-752.190}} \quad (\lambda = 2\text{--}14 \mu\text{m}) \quad [34]$$

$$n_2 = 4.0 \times 10^{-17} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [26]$$

IRG22 (AMTIR1)

$$n_0 = \sqrt{3.4834 + \frac{2.8203\lambda^2}{\lambda^2-0.1352} + \frac{0.9773\lambda^2}{\lambda^2-1420.7}} \quad (\lambda = 0.8\text{--}15.5 \mu\text{m}) \quad [35]$$

$$n_2 = 1.4 \times 10^{-18} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [26]$$

IRG24

$$n_0 = \sqrt{3.8965 + \frac{2.9567\lambda^2}{\lambda^2-0.1620} + \frac{0.9461\lambda^2}{\lambda^2-1939.1}} \quad (\lambda = 0.8\text{--}15.5 \mu\text{m}) \quad [36]$$

$$n_2 = 2.5 \times 10^{-18} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [26]$$

IRG25

$$n_0 = \sqrt{3.7574 + \frac{3.0990\lambda^2}{\lambda^2-0.1596} + \frac{1.6660\lambda^2}{\lambda^2-2045.5}} \quad (\lambda = 0.8\text{--}15.5 \mu\text{m}) \quad [37]$$

$$n_2 = 2.3 \times 10^{-18} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [26]$$

KBr

$$n_0 = \sqrt{1.39408 + \frac{0.79221\lambda^2}{\lambda^2-0.146^2} + \frac{0.01981\lambda^2}{\lambda^2-0.173^2} + \frac{0.15587\lambda^2}{\lambda^2-0.187^2} + \frac{0.17673\lambda^2}{\lambda^2-60.61^2} + \frac{2.06217\lambda^2}{\lambda^2-87.72^2}} \quad (\lambda = 0.2\text{--}42 \mu\text{m}) \quad [32]$$

$$n_2 = 4.3 \times 10^{-20} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [26]$$

KCl

$$n_0 = \sqrt{1.26486 + \frac{0.30523\lambda^2}{\lambda^2-0.100^2} + \frac{0.41620\lambda^2}{\lambda^2-0.131^2} + \frac{0.18870\lambda^2}{\lambda^2-0.162^2} + \frac{2.6200\lambda^2}{\lambda^2-70.42^2}} \quad (\lambda = 0.18\text{--}35 \mu\text{m}) \quad [32]$$

$$n_2 = 3.4 \times 10^{-20} \text{ m}^2/\text{W} \text{ at } 9.2 \mu\text{m} \quad [29]$$

KRS5

$$n_0 = \sqrt{1 + \frac{1.8293958\lambda^2}{\lambda^2 - 0.0225} + \frac{1.6675593\lambda^2}{\lambda^2 - 0.0625} + \frac{1.1210424\lambda^2}{\lambda^2 - 0.1225} + \frac{0.04513366\lambda^2}{\lambda^2 - 0.2025} + \frac{12.380234\lambda^2}{\lambda^2 - 27089.737}} \quad (\lambda = 0.577\text{--}39.4\text{ }\mu\text{m}) \quad [38]$$

$$n_2 = 9.0 \times 10^{-19} \text{ m}^2/\text{W} \text{ at } 9.2\text{ }\mu\text{m} \quad [26]$$

NaCl

$$n_0 = \sqrt{1.00055 + \frac{0.19800\lambda^2}{\lambda^2 - 0.050^2} + \frac{0.48398\lambda^2}{\lambda^2 - 0.100^2} + \frac{0.38696\lambda^2}{\lambda^2 - 0.128^2} + \frac{0.25998\lambda^2}{\lambda^2 - 0.158^2} + \frac{0.08796\lambda^2}{\lambda^2 - 40.50^2} + \frac{3.17064\lambda^2}{\lambda^2 - 60.98^2} + \frac{0.30038\lambda^2}{\lambda^2 - 120.34^2}} \quad (\lambda = 0.2\text{--}30\text{ }\mu\text{m}) \quad [32]$$

$$n_2 = 3.5 \times 10^{-20} \text{ m}^2/\text{W} \text{ at } 9.2\text{ }\mu\text{m} \quad [29]$$

NaF

$$n_0 = \sqrt{1.41572 + \frac{0.32785\lambda^2}{\lambda^2 - 0.117^2} + \frac{3.18248\lambda^2}{\lambda^2 - 40.57^2}} \quad (\lambda = 0.15\text{--}17\text{ }\mu\text{m}) \quad [32]$$

$$n_2 = 6.0 \times 10^{-21} \text{ m}^2/\text{W} \text{ at } 9.2\text{ }\mu\text{m} \text{ at } 9.2\text{ }\mu\text{m} \quad [26]$$

$$\alpha_0 = 5.0(e^{0.97(\lambda-8)} - 1) \text{ m}^{-1} \quad [26]$$

Si

$$n_0 = 3.41983 + \frac{0.159906}{\lambda^2 - 0.028} - 0.123109 \left(\frac{1}{\lambda^2 - 0.028} \right)^2 + 1.26878 \times 10^{-6} \lambda^2 - 1.95104 \times 10^{-9} \lambda^4 \quad (\lambda = 2.44\text{--}25\text{ }\mu\text{m}) \quad [39]$$

$$n_2 = 1.2 \times 10^{-17} \text{ m}^2/\text{W} \text{ at } 9.2\text{ }\mu\text{m} \quad [26]$$

SiO₂

$$n_0 = \sqrt{1 + \frac{0.6961663\lambda^2}{\lambda^2 - 0.0684043^2} + \frac{0.4079426\lambda^2}{\lambda^2 - 0.1162414^2} + \frac{0.8974794\lambda^2}{\lambda^2 - 9.896161^2}} \quad (\lambda = 0.21\text{--}6.7\text{ }\mu\text{m}) \quad [40]$$

$$n_2 = 3.29 \times 10^{-20} \text{ m}^2/\text{W} \text{ at } 1.06\text{ }\mu\text{m} \quad [31]$$

ZnS

$$n_0 = \sqrt{8.393 + \frac{0.14383}{\lambda^2 - 0.2421^2} + \frac{4430.99}{\lambda^2 - 36.71^2}} \quad (\lambda = 0.405\text{--}13\text{ }\mu\text{m}) \quad [41]$$

$$n_2 = 4.0 \times 10^{-19} \text{ m}^2/\text{W} \text{ at } 9.2\text{ }\mu\text{m} \quad [26]$$

ZnSe

$$n_0 = \sqrt{1 + \frac{4.45813734\lambda^2}{\lambda^2 - 0.200859853^2} + \frac{0.467216334\lambda^2}{\lambda^2 - 0.391371166^2} + \frac{2.89566290\lambda^2}{\lambda^2 - 47.1362108^2}} \quad (\lambda = 0.54\text{--}18.2\text{ }\mu\text{m}) \quad [42]$$

$$n_2 = 6.5 \times 10^{-19} \text{ m}^2/\text{W} \text{ at } 9.2\text{ }\mu\text{m} \quad [26]$$

Appendix D

Selected formulas and methods explained

D.1 Equation 4.10

Eq. 4.10 defines the fraction z_{jk} of discharge energy spent in inelastic collisions:

$$z_{jk} = 10^{16} \frac{y_j u_{jk} \omega_{jk}}{\left(\frac{\xi \mathcal{E}}{\mathcal{N}} \right) v_d}$$

where $y_j[-]$ is the relative concentration of a component in the gas mixture, $u_{jk}[\text{eV}]$ is the transferred energy per electron-molecule collision, collision rate constant $\omega_{jk}[\text{cm}^3 \text{s}^{-1}]$ divided by electron drift speed $v_d[\text{cm}^3 \text{s}^{-1}]$ is the collision cross-section ($[\text{cm}^2]$), $\mathcal{E}[10^{-16} \text{V cm}^{-1}]$ is the electric field, $\xi[\text{eV V}^{-1}]$ is the energy gained by electron moved across an electric potential difference of 1 V, and $\mathcal{N}[\text{cm}^{-3}]$ is the total absolute concentration of the gas mixture.

The physical meaning of $\xi \mathcal{E}$ is the energy (in eV) gained by an electron after passing 1 cm in the electric field $\mathcal{E}$. By definition of electronvolt, $\xi = 1$ and is thus omitted in Eq. 4.10.

D.2 Pumping rate constants in equations 4.13

Pumping rate constant is the number of quanta added to a given vibrational mode per unit of time per molecule.

$$p_e = \frac{1}{E_v[\text{J}]} \times \frac{1}{N[\text{cm}^{-3}]n[-]y[-]} \times q[-]W[\text{J s}^{-1} \text{cm}^{-3}]$$

where E_v is the energy of the vibrational quanta: $4.665 \times 10^{-20} \text{J}$ (2349cm^{-1}) for ν_3 mode of CO_2 (and roughly same for N_2 vibration), and $1.325 \times 10^{-20} \text{J}$ (667cm^{-1}) for ν_2 mode; $N = 2.7 \times 10^{19} \text{cm}^{-3}$ is the density of gas molecules under normal conditions (1 bar, 273 K); q is the fraction of discharge energy deposited in the corresponding vibration; n is the correction factor for molecular density at the conditions different from 'normal'; y is the relative concentration of the gas in the mixture; W is the discharge power density.

Combining the constants and switching to kW/cm^3 for power density and μs^{-1} for the rate constants we get the formulas given in the model description:

$$p_{e4} = 0.8 \times 10^{-3} \frac{q_4}{ny_2} W(t); \quad p_{e3} = 0.8 \times 10^{-3} \frac{q_3}{ny_1} W(t); \quad p_{e2} = 2.8 \times 10^{-3} \frac{q_2}{ny_1} W(t);$$

D.3 Solution of Boltzmann equation 4.7

To find the distribution function $f(u)$ from Eq. 4.7, we first represent the equation in the following discrete form.

$$\begin{aligned}
& \frac{1}{3} \left(\frac{\mathcal{E}}{\mathcal{N}} \right)^2 \frac{1}{2du} \left[u_{i+1} \left(\sum_j y_j Q_{mj}(u_{i+1}) \right)^{-1} \frac{f_{i+2} - f_i}{2du} - u_{i-1} \left(\sum_j y_j Q_{mj}(u_{i-1}) \right)^{-1} \frac{f_i - f_{i-2}}{2du} \right] \\
& + 1.09 \times 10^{-3} \frac{1}{2du} \left[u_{i+1}^2 f_{i+1} \sum_j \frac{y_j}{M_j} Q_{mj}(u_{i+1}) - u_{i-1}^2 f_{i-1} \sum_j \frac{y_j}{M_j} Q_{mj}(u_{i-1}) \right] \\
& + \sum_{j=1,2} y_j C_j \frac{u_{i+1} f_{i+1} - u_{i-1} f_{i-1}}{2du} + 6B y_2 \frac{u_{i+1} Q(u_{i+1}) f_{i+1} - u_{i-1} Q(u_{i-1}) f_{i-1}}{2du} (u Q(u) f) \\
& + \sum_j y_j \sum_k (u_i + u_{jk}) Q_{jk}(u_i + u_{jk}) f_x - u_i f_i \sum_j y_j \sum_k Q_{jk}(u_i) = 0,
\end{aligned}$$

where du is the grid step, $u_i = i \times du$, $f_i = f(u_i)$, and $f_x = f(i + u_{jk}/du)$. We use 1024 steps in the grid $i = 0 \dots 1023$.

We now have a system of 1024 equations for 1024 values of f_i corresponding to the values of u_i . The coefficients of the equations are summarized below.

$\mathbf{f}_i :$

$$\begin{aligned}
& - \sum_j y_j \sum_k u_i Q_{jk}(u_i) - \\
& - \frac{1}{3} \left(\frac{\mathcal{E}}{\mathcal{N}} \right)^2 \frac{1}{4(du)^2} \left[u_{i+1} \left(\sum_j y_j Q_{mj}(u_{i+1}) \right)^{-1} - u_{i-1} \left(\sum_j y_j Q_{mj}(u_{i-1}) \right)^{-1} \right]
\end{aligned}$$

$\mathbf{f}_{i+u_{jk}/du} :$

$$y_j (u_i + u_{jk}) Q_{jk}(u_i + u_{jk})$$

$\mathbf{f}_{i+1} :$

$$1.09 \times 10^{-3} \frac{1}{2du} u_{i+1}^2 f_{i+1} \sum_j \frac{y_j}{M_j} Q_{mj}(u_{i+1}) + \sum_{j=1,2} y_j C_j \frac{u_{i+1}}{2du} + 6B y_2 \frac{u_{i+1} Q(u_{i+1})}{2du}$$

$\mathbf{f}_{i-1} :$

$$- 1.09 \times 10^{-3} \frac{1}{2du} u_{i-1}^2 f_{i-1} \sum_j \frac{y_j}{M_j} Q_{mj}(u_{i-1}) - \sum_{j=1,2} y_j C_j \frac{u_{i-1}}{2du} - 6B y_2 \frac{u_{i-1} Q(u_{i-1})}{2du}$$

$\mathbf{f}_{i+2} :$

$$\frac{1}{3} \left(\frac{\mathcal{E}}{\mathcal{N}} \right)^2 \frac{1}{4(du)^2} u_{i+1} \left(\sum_j y_j Q_{mj}(u_{i+1}) \right)^{-1}$$

$\mathbf{f}_{i-2} :$

$$\frac{1}{3} \left(\frac{\mathcal{E}}{\mathcal{N}} \right)^2 \frac{1}{4(du)^2} u_{i-1} \left(\sum_j y_j Q_{mj}(u_{i-1}) \right)^{-1}$$

These equations are then solved using the tridiagonal matrix algorithm.

D.4 Solution of the polarization equation 4.19

In the amplification model, the polarization $\rho_j(t)$ for each ro-vibrational transition j obeys the linear first-order equation (Eq. 4.19):

$$\frac{\partial \rho_j}{\partial t} + \left(2\pi i(\nu_c - \nu_j) + \frac{1}{\tau_2} \right) \rho_j = -\frac{\sigma_j n_j E}{2\tau_2}. \quad (\text{D.1})$$

During one time step Δt of the pulse time grid, the code updates ρ_j by treating the complex coefficient multiplying ρ_j and the driving term as constant over the step. Specifically, for the step $t^n \rightarrow t^{n+1} = t^n + \Delta t$, the field in the drive is taken as the field *before* amplification at that time sample, denoted E_{in} in the code, and the population inversion n_j is taken from the current populations at the same step.

With these assumptions, over a single step the equation becomes an inhomogeneous linear ODE with constant coefficients. Dropping the subscript j for brevity,

$$\frac{d\rho}{dt} = -a\rho + b, \quad (\text{D.2})$$

where

$$a = \frac{1}{\tau_2} + i 2\pi(\nu_c - \nu_j), \quad b = -\frac{\sigma_j n_j}{2\tau_2} E_{\text{in}}. \quad (\text{D.3})$$

The exact one-step update is

$$\rho^{n+1} = \rho^n e^{-a\Delta t} + b \frac{1 - e^{-a\Delta t}}{a}. \quad (\text{D.4})$$

This can be viewed as an integrating-factor method: the factor $e^{-a\Delta t}$ accounts exactly for dephasing ($1/\tau_2$) and detuning ($\nu_c - \nu_j$), while the second term accounts for the (frozen) drive over the same interval.

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